Basic Principles of Medicine 1 Module: Foundations to Medicine Lecture 17

Lecture Title: Molecular Biology Techniques

Name of Lecturer: Dr. Mary Maj, PhD
St Georges University ©

Pre-Lecture - Required Reading and Objectives

- Textbook: Biochemistry, Lippincott Illustrated Reviews, 8th Edition, Chapter 34, Biotechnology and Human Disease
  - Link: Biotechnology and Human Disease | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library (lwwhealthlibrary.com)

Objectives

SOM.MK.I.BPM1.1.FTM.3.GNET.0200	Explain the general principles and utilization of various molecular biology tools and techniques, including Restriction endonucleases, DNA cloning, Probes, Southern Blotting, Restriction Fragment Length Polymorphisms, Polymerase chain reaction and Gene Expression analysis
SOM.MK.I.BPM1.1.FTM.3.GNET.0201	Summarize how complimentary DNA (cDNA) is generated from messenger RNA (mRNA) and how this technique is used to express human proteins in a bacteria to generate therapeutic enzymes such as insulin

Recommended Reading

- You can find these chapters on your Sakia site under:
  - Resources/General Information/Book Access or from lessons page
- Textbook Human and Genetics and Genomics, 4th Edition
  - Chapter 4, page 68 , figure 4.3 which shows DNA sequencing with fluorescently labeled dideoxynucleotides.
  - Chapter 10, Part I, II, III, and IV of the Neurofibromatosis case (these are in green boxes) which uses sequencing for diagnosis. Also, pay attention to Table 10.2 and Figure 10.3, 10.4, 10.5.
  - Chapter 10, page 178-179, section: Molecular Diagnosis of Genetic Disorders and the accompanying figure 10.6

Molecular Diagnosis: Objectives for Lecture and DLAs

SOM.MK.I.BPM1.1.FTM.3.GNET.0200 (DLA)	Explain the general principles and utilization of various molecular biology tools and techniques, including Restriction endonucleases, DNA cloning, Probes, Southern Blotting, Restriction Fragment Length Polymorphisms, Polymerase chain reaction and Gene Expression analysis
SOM.MK.I.BPM1.1.FTM.3.GNET.0201 (DLA)	Summarize how complementary DNA (cDNA) is generated from messenger RNA (mRNA) and how this technique is used to express human proteins in a bacteria to generate therapeutic enzymes such as insulin
SOM.MK.I.BPM1.1.FTM.3.GNET.0202	Identify the differences between mutations which affect the structures of a gene (or gene region), the structure of the transcribed mRNA product and the structure of the translated protein
SOM.MK.I.BPM1.1.FTM.3.GNET.0203	Explain the concept of hybridization: between a complementary single stranded DNA probe and its target (DNA or RNA); between an antibody and its protein target
SOM.MK.I.BPM1.1.FTM.3.GNET.0204	Compare data from RFLP, ASO, DNA sequencing: from normal homozygotes, carriers (heterozygotes), and affected homozygotes
SOM.MK.I.BPM1.1.FTM.3.GNET.0205	Predict the application of molecular biology techniques for detection of a single base mutation versus small insertion or deletion of DNA sequence, versus large insertion or deletion of DNA sequence.

Molecular Biology

Gene

Techniques to analyze genetic variants

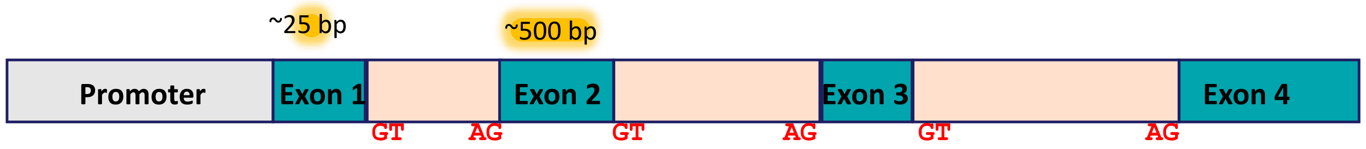
- Insertions or deletions of genetic material
  - In a gene → deletion or insertion of genetic information that alters the gene product
  - In a promoter → the gene will not be expressed
  - In an exon → reading frame altered = wrong amino acids, wrong folding, usually premature stop codon as well = polypeptide that is not a protein or enzyme

- Base pair changes:
  - In promoter region of a gene → no mRNA is made
  - In exons → amino acid will be changed = altered or defective enzyme activity
  - In introns → spliceosome may not form, intron retained affecting coding frame
  - Acceptor site/Donor site mutations → defective splicing & mRNA is changed & may introduce premature stop codon = polypeptide that is not a protein or enzyme
- Polymorphisms:
  - Variants that are seen in greater than 1% of the population, generally not disease causing
  - A polymorphism may be LINKED to a disease-causing mutation

(see Korf and Irons, chapter 10, page 179, figure 10.8)

What is our substrate for analysis?

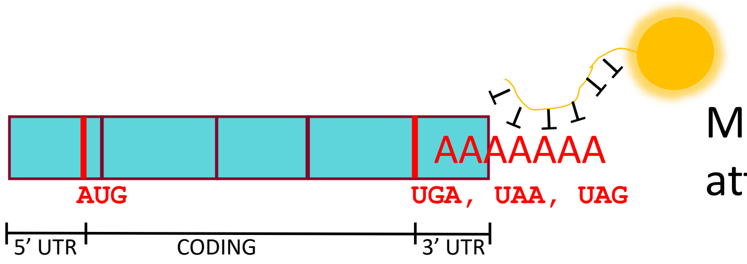
Gene



Genome must be fragmented by cutting with restriction endonucleases which cut at specific sequences in the DNA (in a reproducible manner) and the fragments separated by size by electrophoresis.

- Probe is designed to identify changes in size of fragment of interest
- Insertions or deletions mainly

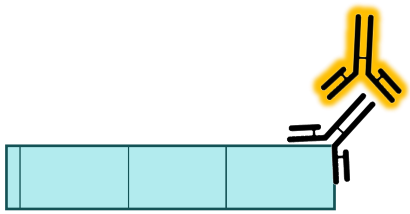
Mature mRNA



Mature mRNA can be isolated from cells by hybridization of poly T tail attached to bead, then run on gel to separate by size

- Probe is designed to identify size differences in mRNA of interest
- Splice site or promoter mutations mainly

Protein



Proteins of interest in cell lysate, separated by size with electrophoresis

- Antibody is designed to hybridize to amino acid sequence in protein of interest
- Secondary antibody designed to hybridize first antibody
- Premature stop codons mainly, exon skipping only if in frame
  - (if amino acids change, the first antibody will not recognize nor hybridize)

From the study DNA in the lab to diagnostics

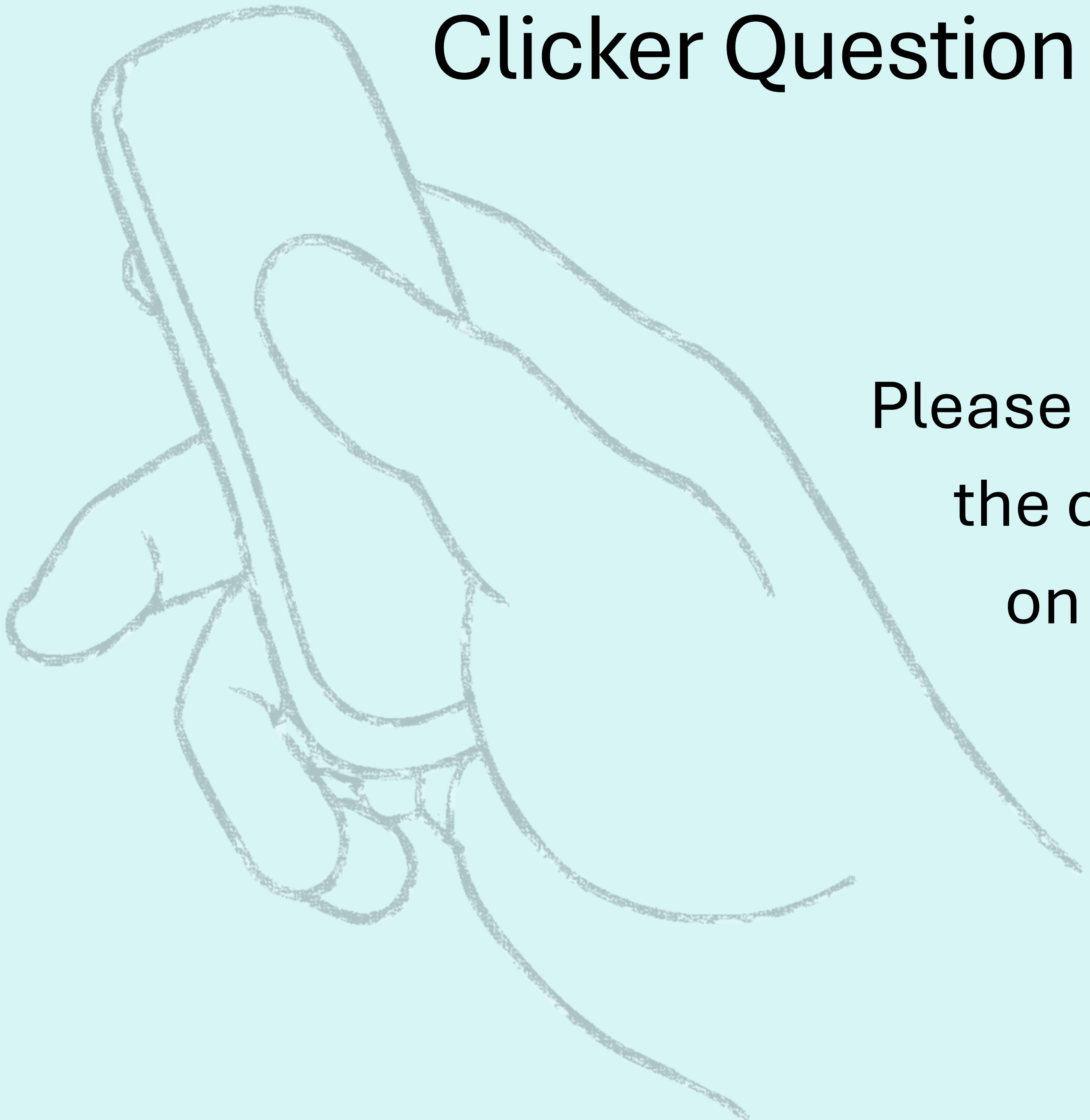
- **Enzymes** which predictably manipulate DNA
  - Restriction endonucleases, ligases & polymerases
- **Cloning** of target DNA into a vector
  - Generate unlimited amounts of target DNA
- **Polymerase chain reaction** (PCR)
  - Alternative to cloning to generate unlimited DNA of interest
  - Designing a PCR Primer

Included in the Assigned Reading

- **Hybridization**: Annealing of ssDNA probe to complementary ssDNA
  - Southern blots
  - Northern blots (RNA)
  - RFLP analysis (Restriction fragment length polymorphism)
  - Allele specific oligonucleotide (ASO probes)
  - Sequencing (primer)
  - Western blots (Protein)

This Lecture

Clicker Question Next!



Please prepare to answer
the clicker question
on the next slide

Testing Genomic DNA – Southern Blot

- 23 chromosome pairs, total of 3.2 billion base pairs
- Chromosomes are large and difficult to study
- DNA is better managed if cut into fragments
- Detection of insertions and deletions by cleaving the DNA into fragments in a reproducible manner
  - Restriction endonucleases which cleave at specific sites on the DNA
- Compare a patient’s DNA fragmentation pattern to control DNA
  - Separate the DNA fragments according to size
  - Find a way to identify fragments of interest
- Identify insertions or deletions

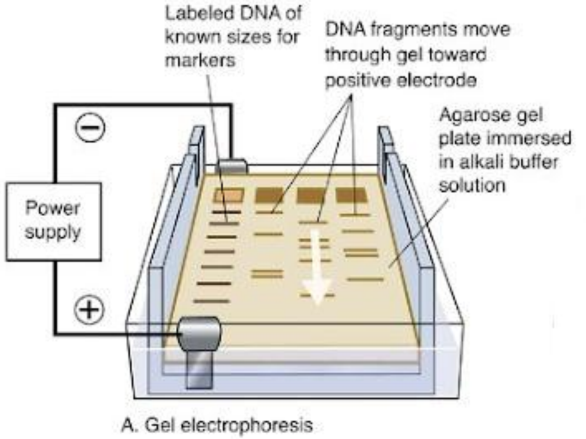
Testing Genomic DNA – Southern Blot

1. Digest DNA with Restriction Endonucleases

- Example: *SmaI* endonuclease recognizes and cleaves DNA at all CCCGGG sequences
- Millions of fragments generated

2. Separate DNA by Size

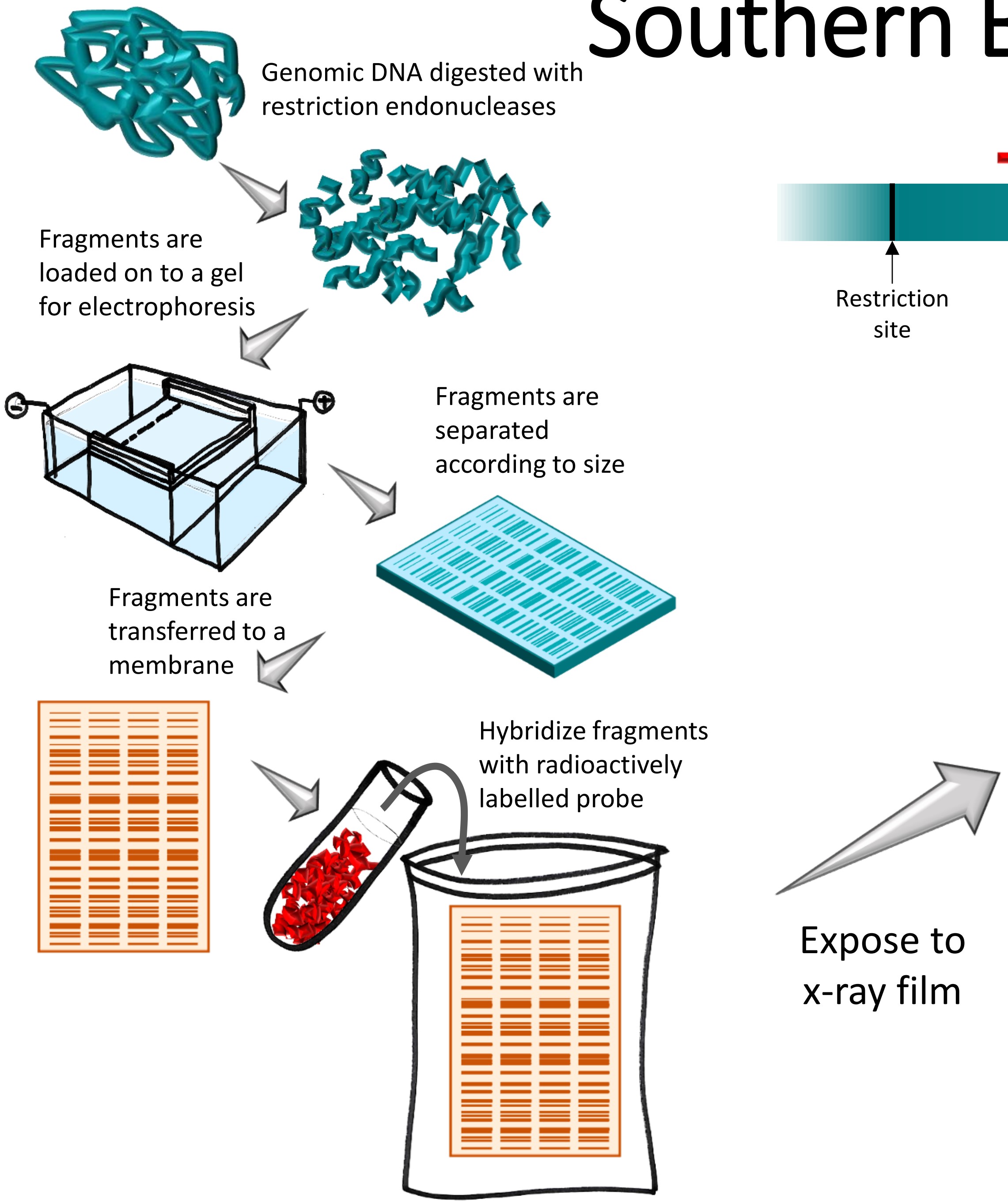
- Fragments separated by gel electrophoresis

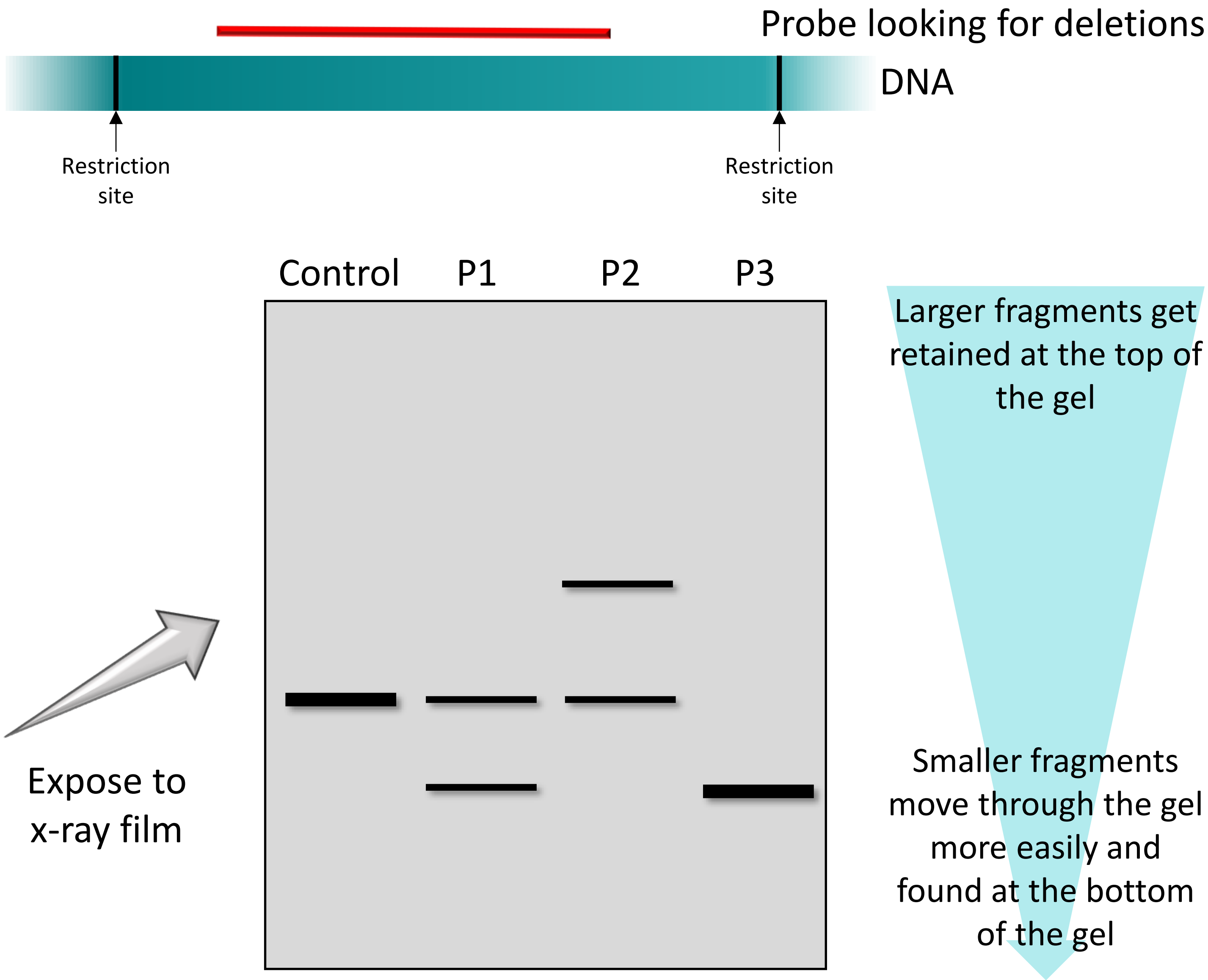


3. Hybridize your region of interest with a probe

- Probe typically synthesized single stranded DNA (ssDNA) that is complimentary to a region of DNA that you are interested in
- Probe is labelled with fluorescent or radioactive tag

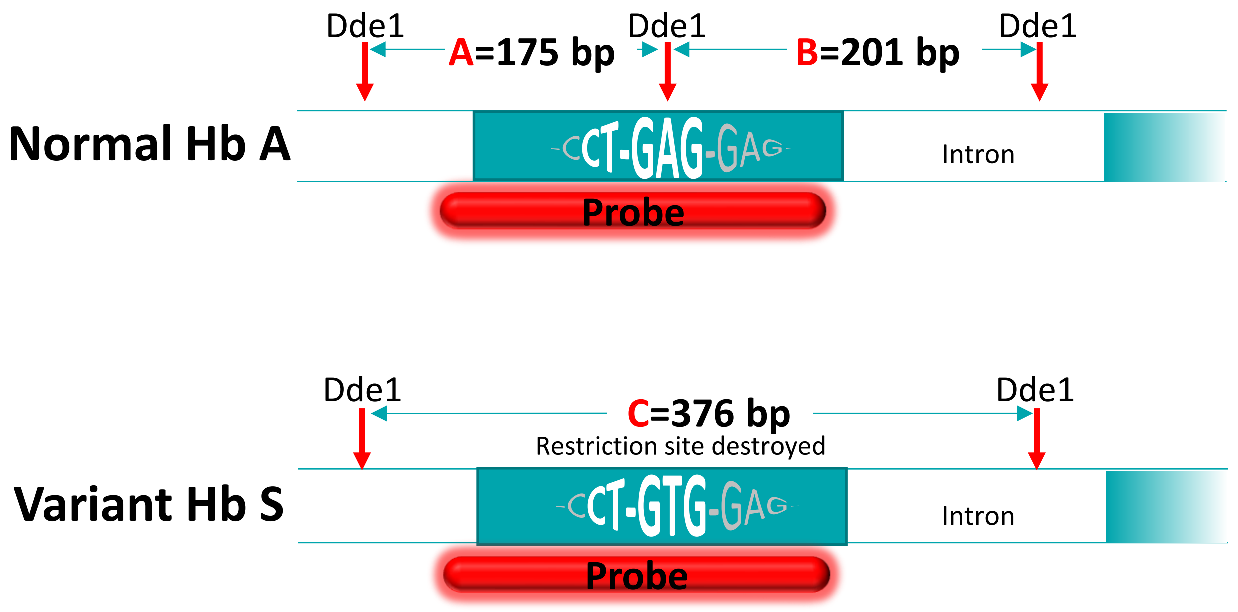
Southern Blots…looking for deletions and insertions

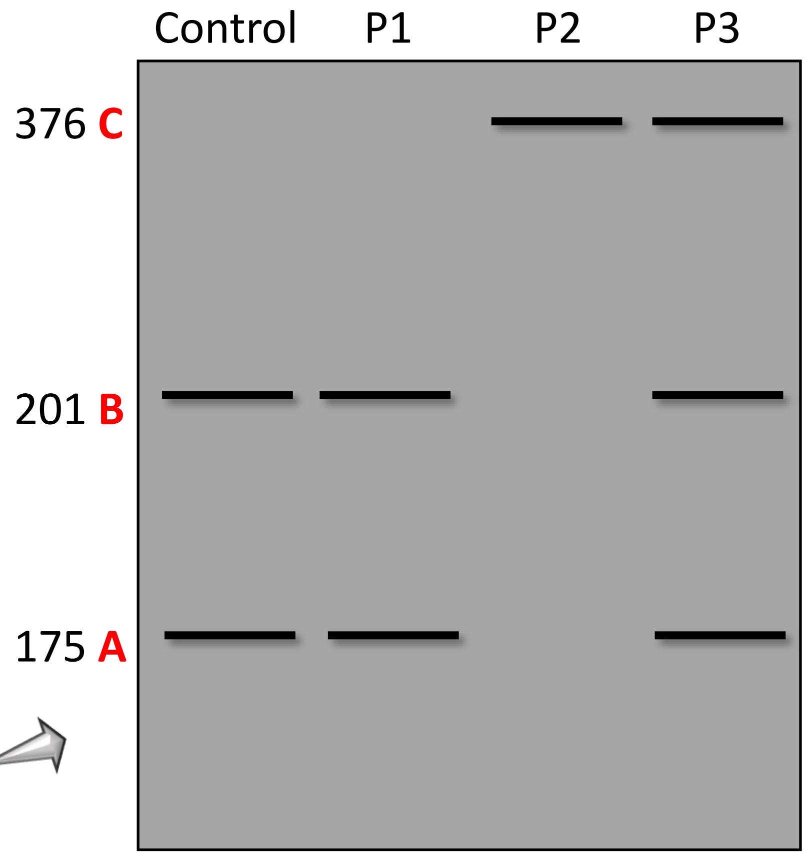


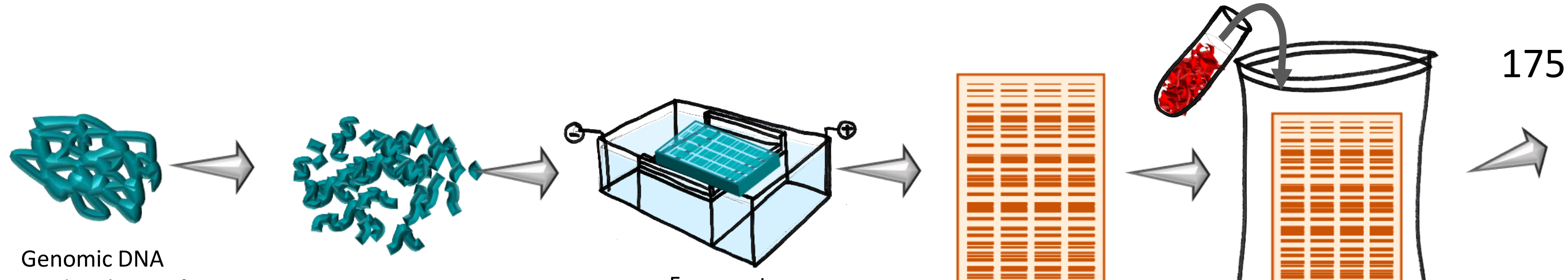


Southern Blot-RFLP: Mutation that creates or destroys a restriction site

Mutation that causes Sickle Cell destroys a restriction site







Genomic DNA digested with specific restriction endonuclease for our test

Fragments are loaded on to a gel for electrophoresis

Fragments are separated according to size

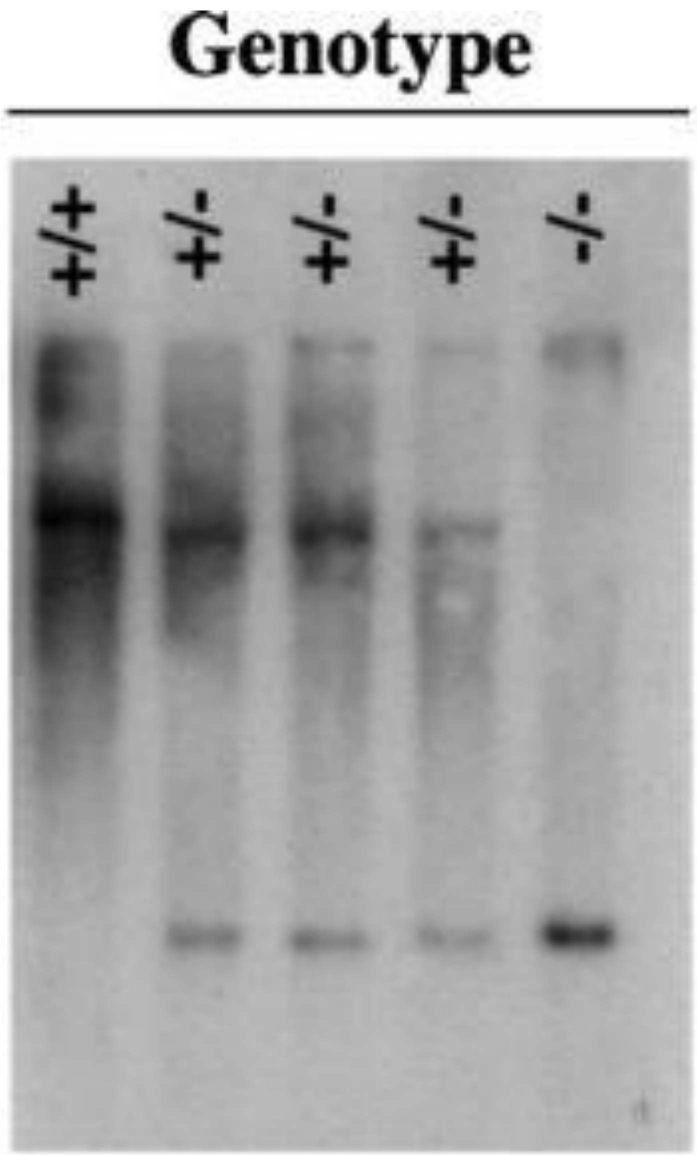
Fragments are transferred to a membrane

Hybridize fragments with radioactively labelled probe

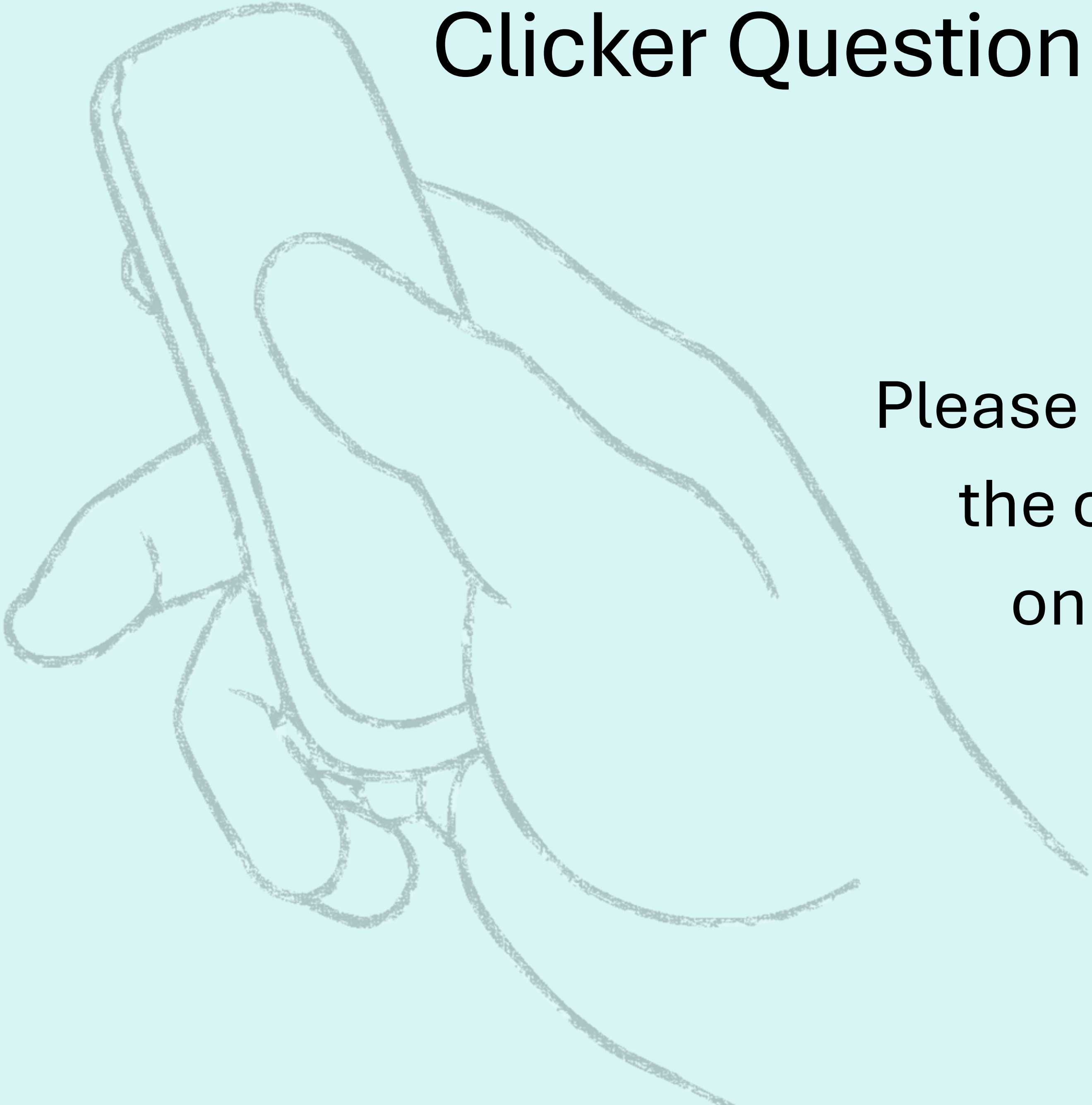
Expose to x-ray film

Southern Blots (words)

- ssDNA probe, complementary to gene region of interest
- Probe tagged with either radioactive isotope or fluorescent marker
  - Presence vs. absence of DNA region
  - Presence vs. absence of point mutation (make/break restriction site)
- DNA purified from cell, cut with restriction enzymes & separated by gel electrophoresis
- Gel is soaked in an alkaline solution (denature DNA)
- DNA transferred from gel to DNA-binding membrane
- Radioactive probe is hybridized to the membrane
- Excess probe washed
- Membrane exposed to X-ray film
- DNA fragments of interest is visualized



Clicker Question Next!



Please prepare to answer
the clicker question
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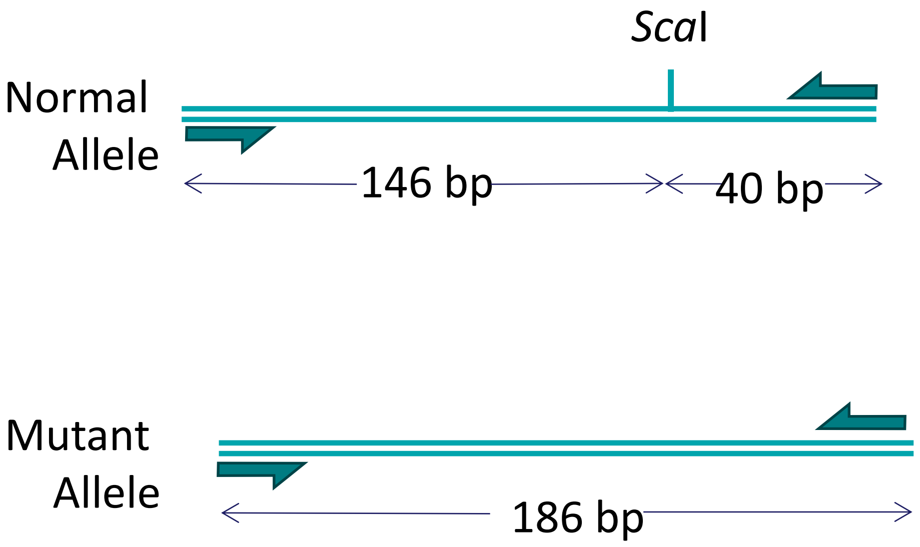
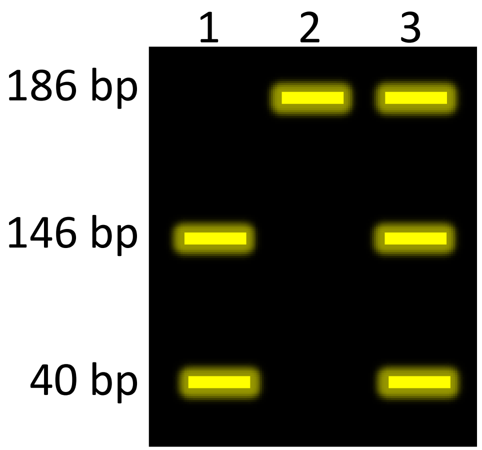
PCR-RFLP Analysis: Mutation that creates or destroys a restriction site

- Diagnosis of Maple Syrup Urine Disease
- Common mutation Y393N which **destroys** a *specific* restriction site (ScaI)
  - Autosomal recessive disease, thus test will reveal carrier status
- **First: PCR** amplify a 186 bp fragment from the MSUD gene
- **Next:** Digest with the appropriate *(this test ScaI)* restriction enzyme and separate DNA fragments according to size by electrophoresis
  - visualize DNA fragments using ethidium bromide and UV light





Allele Specific Oligonucleotide (ASO) or Dot Blot

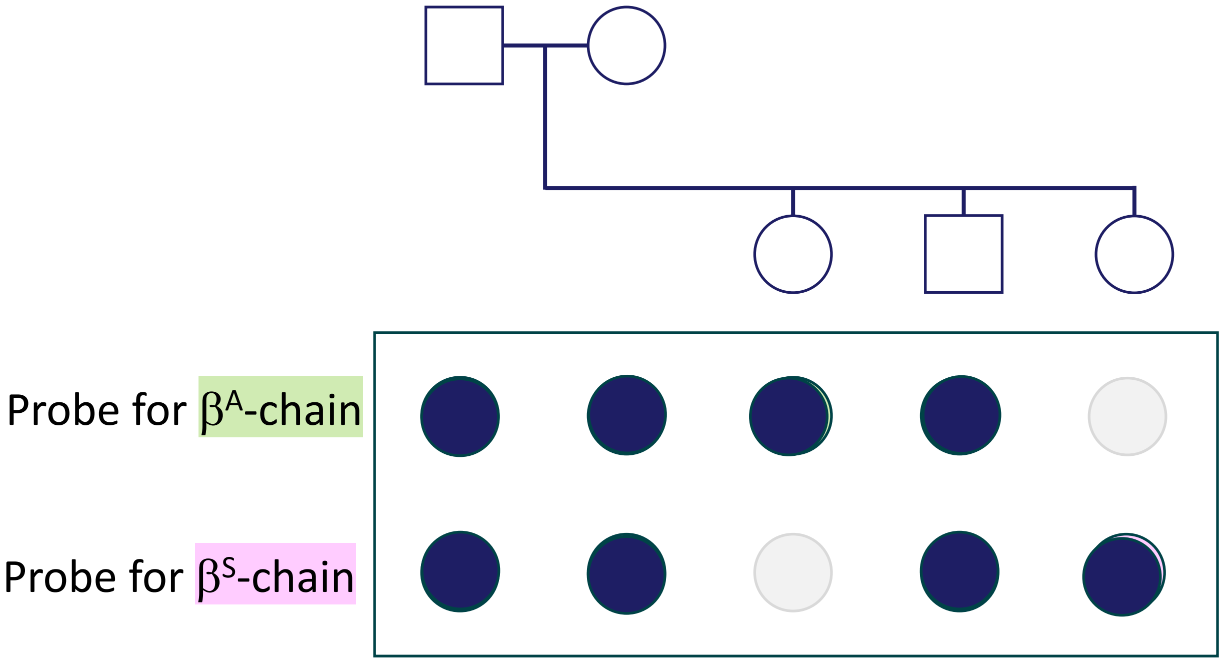
ASO design to genotype Sickle Cell Anemia: missense mutation, no allelic heterogeneity

- Two probes designed:
  - One to bind the normal allele and
  - Other to bind mutant allele
- DNA prep:
  - DNA is isolated from patient
  - Spotted twice on a membrane

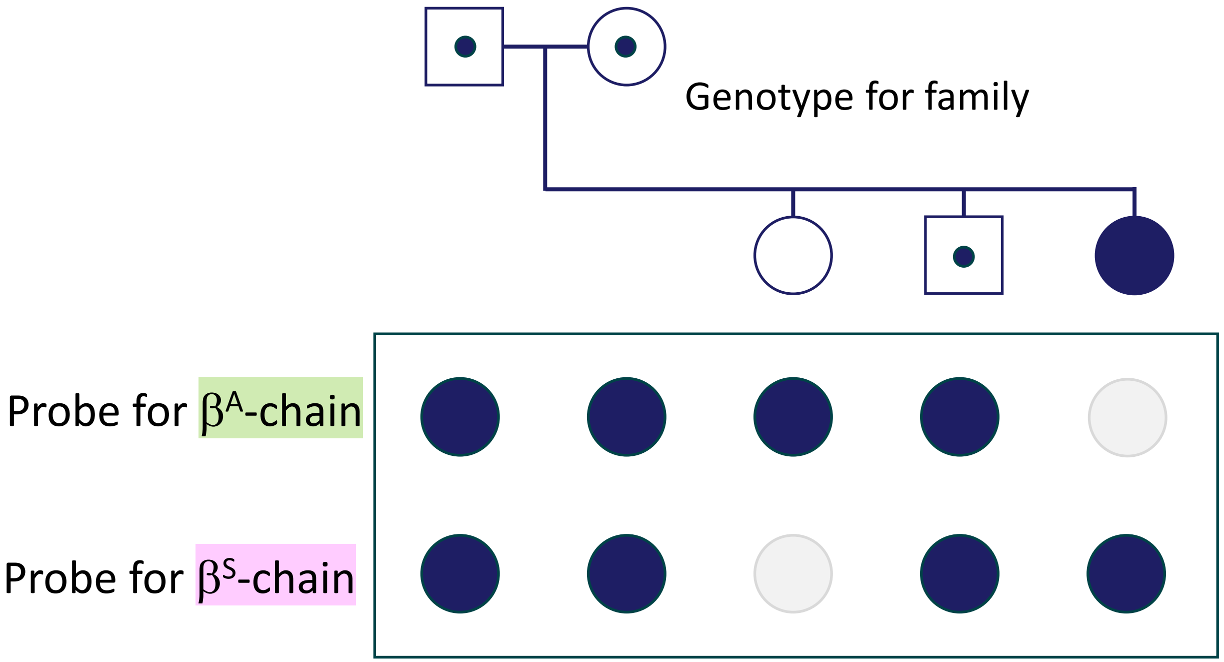
β^A-chain of HbA	ATGGTGCACCTGACTCCTGAGGAGAAGTCT M V H L T P E E K S	ATGGTGCACCTGACTCCTGAGGAGAAGTCT M V H L T P E E K S
β^S-chain of HbS	ATGGTGCACCTGACTCCTGTGGAGAAGTCT M V H L T P V E K S	ATGGTGCACCTGACTCCTGTGGAGAAGTCT M V H L T P V E K S

ASO for Sickle Cell (no allelic heterogeneity)

- This family is affected by Sickle cell anemia
  - DNA isolated for family and blotted twice on membrane
  - Top row incubated with probe for the normal allele
  - Bottom row is incubated with probe for the mutant allele
  - Hybridization occurs when the probe is complementary to DNA
  - Probes are radioactive and visualized by exposing membrane to X-ray film



Probe Hybridization Pattern
False color for demonstration- which probes bind where.



Probe Hybridization Pattern
X-ray film shows where radioactivity is bound

ASO for Cystic Fibrosis: Allelic Heterogeneity

- Mom and dad both have a family history of cystic fibrosis
- Mom's family has the ΔF508 mutation
- Dad's family has the R117H mutation
- Four probes designed to determine carrier status of mom and dad:
  - One to bind normal allele F508
  - One to bind variant allele ΔF508
  - One to bind normal allele R117
  - One to bind variant allele R117H

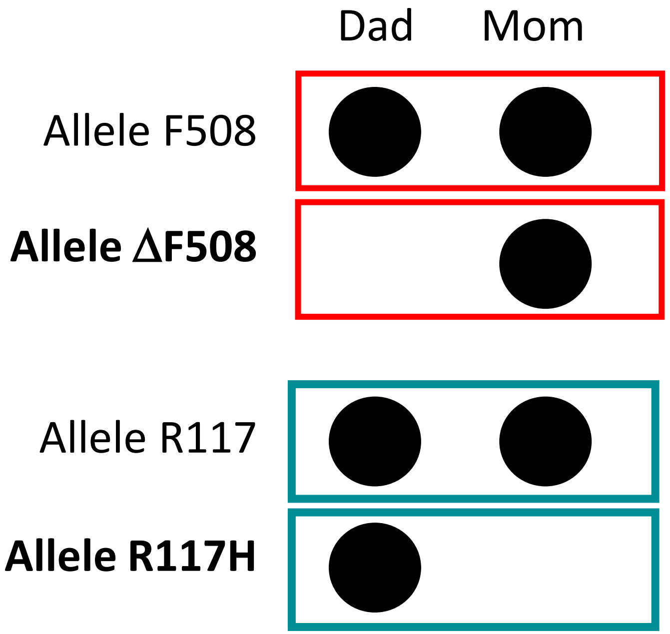
ASO for Cystic Fibrosis: Allelic Heterogeneity

ASO design to genotype cystic fibrosis: deletion mutation and a missense mutation

Allele F508	ATTAAAGAAAATATCATCTTTGGTGTTTCC I K E N I I F G V S 508	ATTAAAGAAAATATCATTGGTGTTTCC I K E N I I **G** V S 508	Allele ΔF508
Allele R117	GATAACAAGGAGGAACGCTCTATCGCGATT D N K E E R S I A V 117	GATAACAAGGAGGAACACTCTATCGCGATT D N K E E H S I A V 117	Allele R117H

ASO Cystic Fibrosis: Parent’s Dot Blot for Genotype

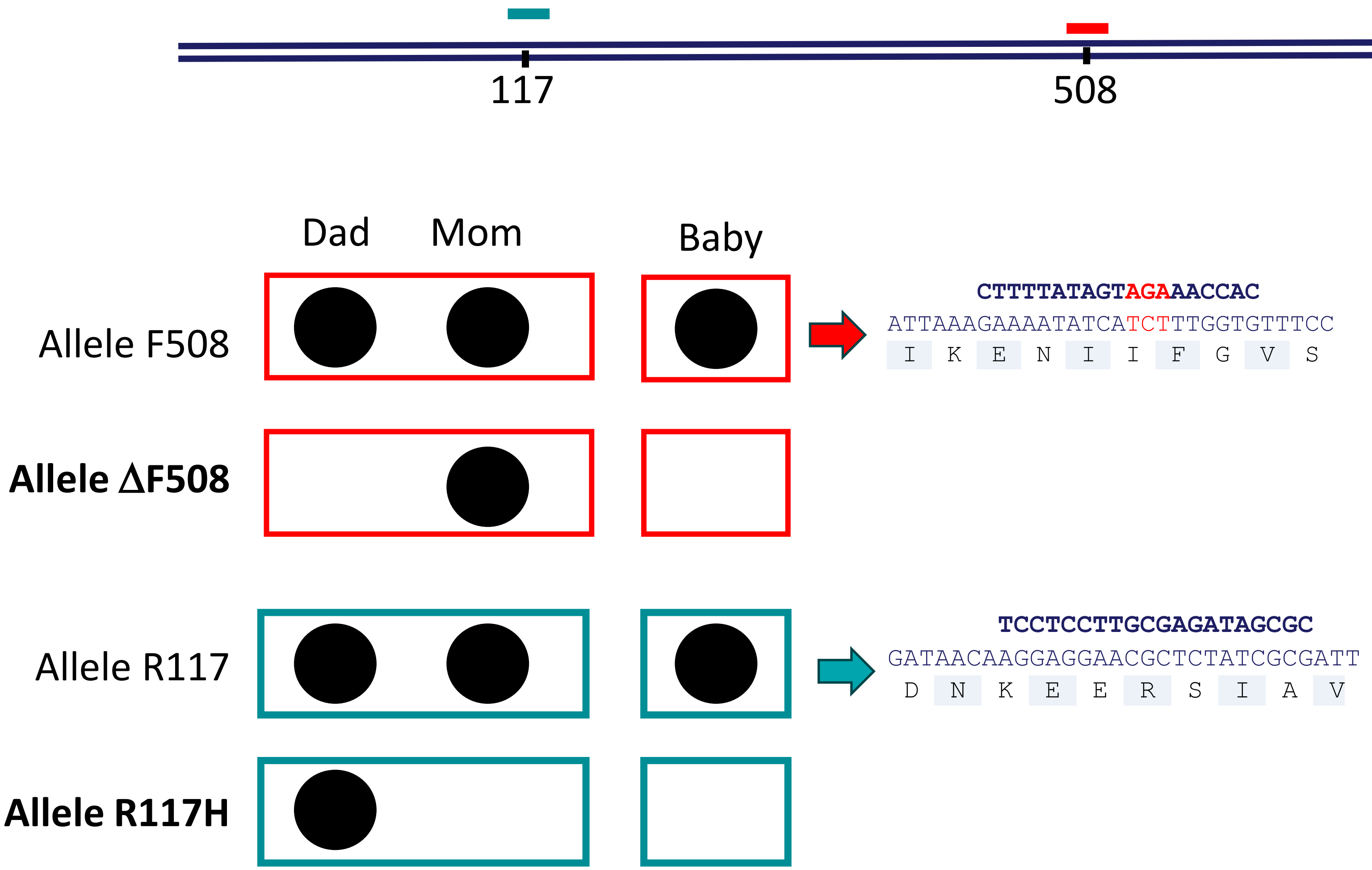
- DNA prep:
  - DNA is isolated from patient
  - Spotted 4 times on a membrane in a column
  - Each row is hybridized with a unique probe



Mom carrier of ΔF508 mutation

Dad carrier of R117H mutation

ASO Cystic Fibrosis: Chance that baby is homozygous normal?

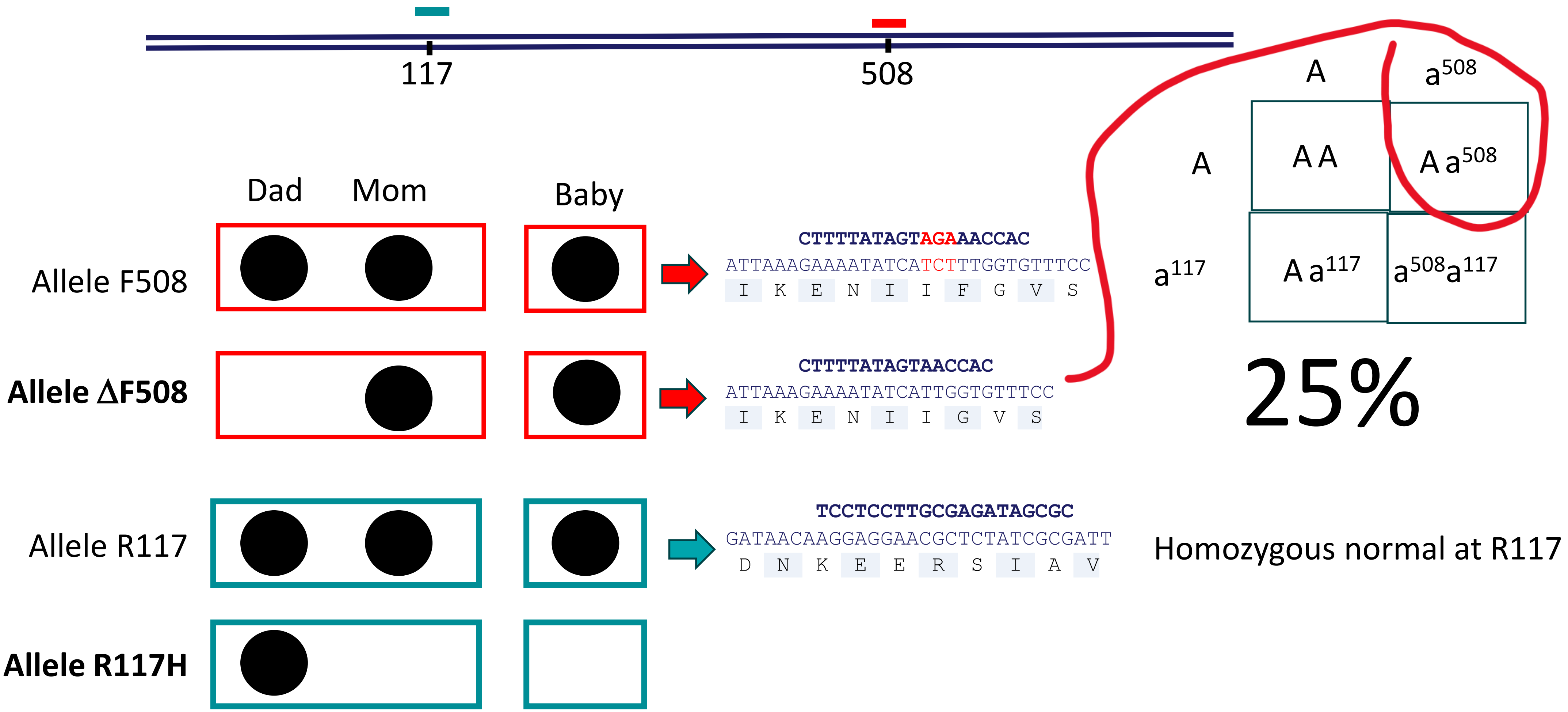


	A	a^{508}
A	A A	A a^{508}
a^{117}	A a^{117}	$a^{508}a^{117}$

25%

Do the Punnet square, baby has 25% of inheriting both normal alleles

ASO Cystic Fibrosis: Chance that baby is carrier of ΔF508?



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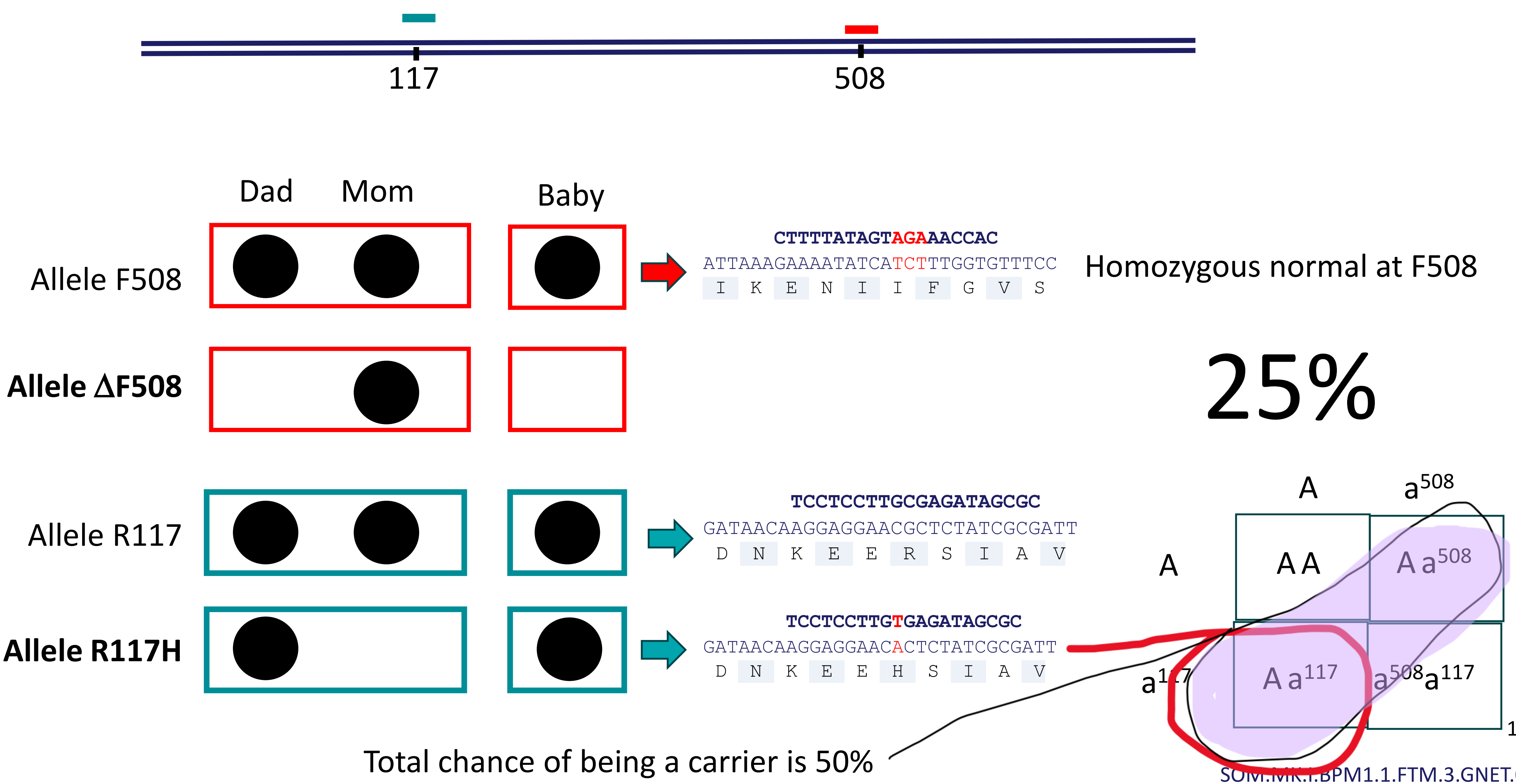
ASO Cystic Fibrosis: Chance that baby is carrier of R117H?



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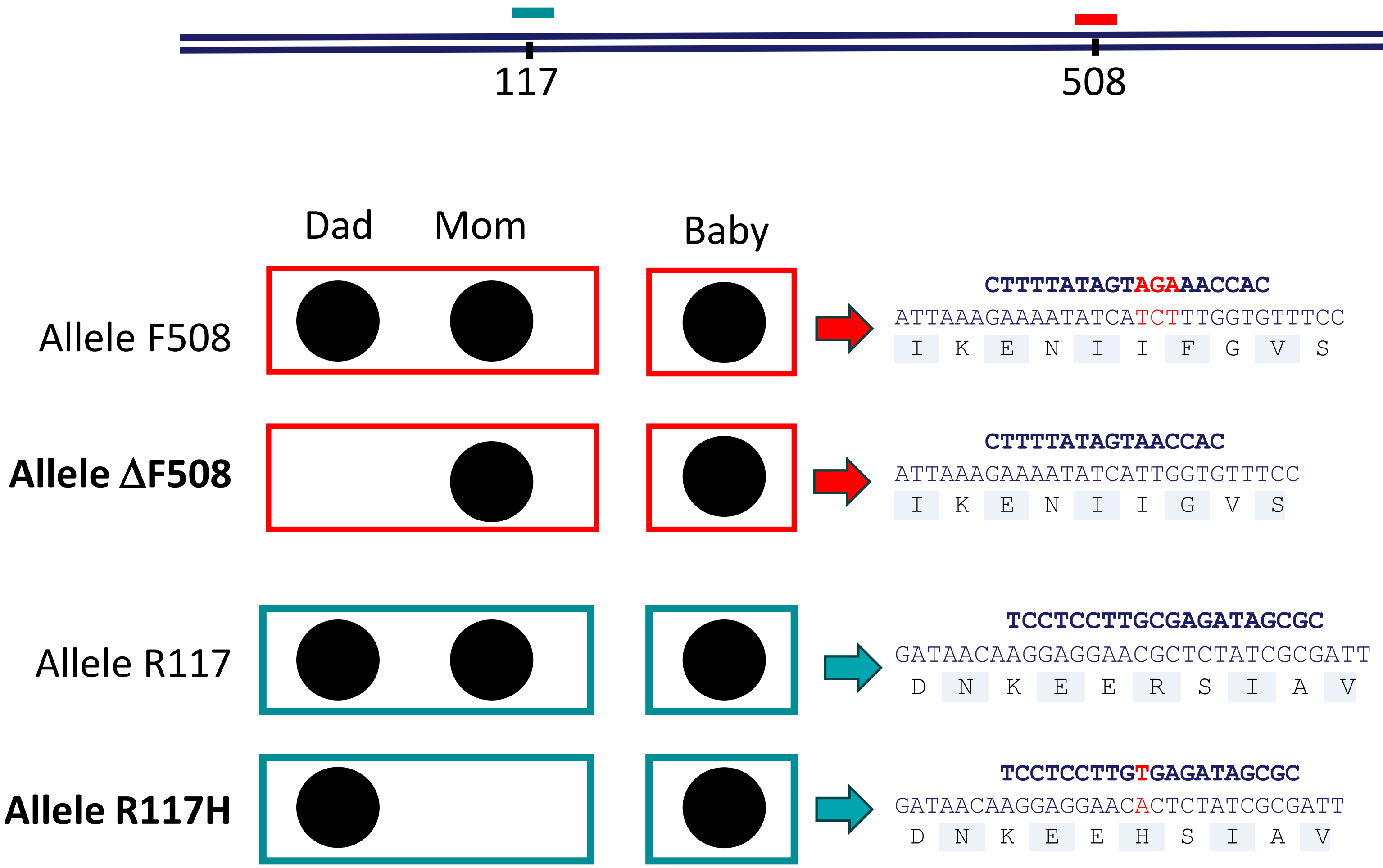
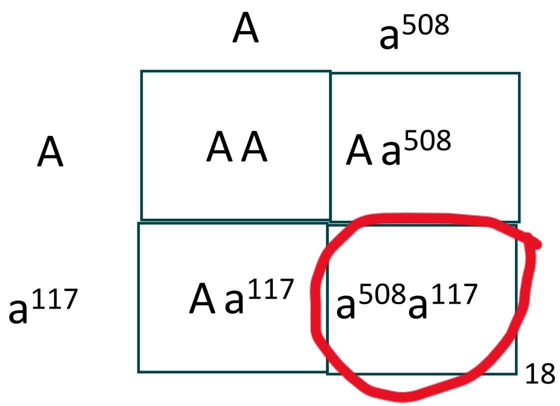
ASO Cystic Fibrosis: Chance that baby has cystic fibrosis?



25%



This scenario has a name.....

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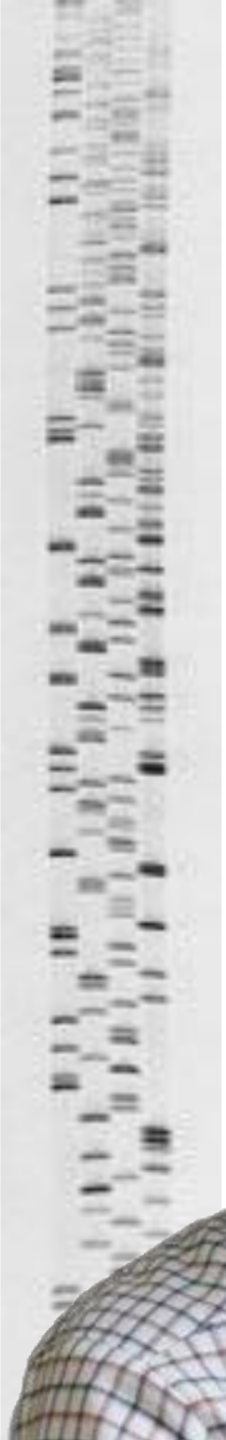
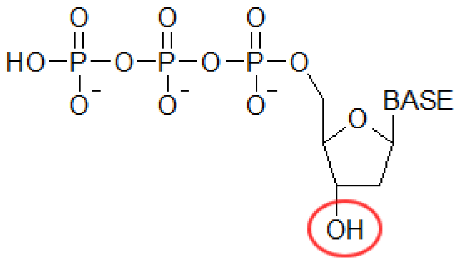
What if a family has cystic fibrosis, but genotype tests are all normal?

DNA sequencing

Sanger Sequencing: di-deoxy method

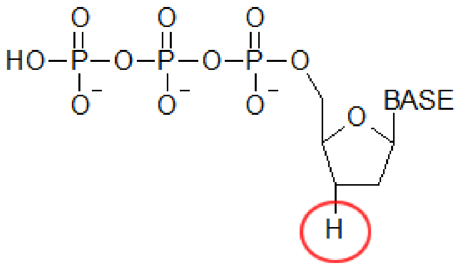
AKA Chain Termination Sequencing



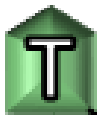


Deoxynucleotide triphosphate (dNTP)

DNA polymerase needs 3’-OH to add the next in-coming nucleotide for synthesis



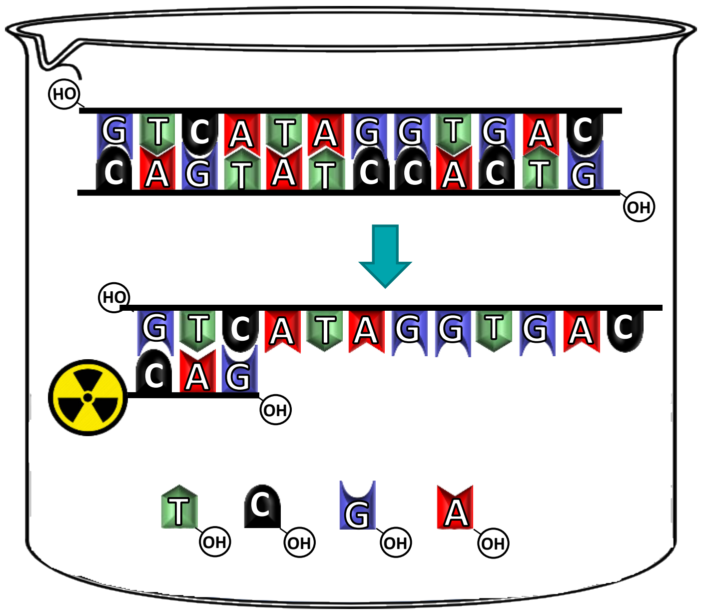
Dideoxynucleotide triphosphate ddNTP



No 3’-OH then synthesis will **terminate**

Fredrick Sanger

Sanger Sequencing: Sequencing reaction mixture



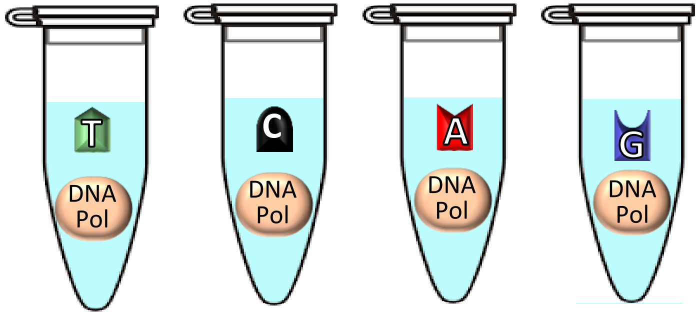
1. DNA to be sequenced

2. Denature to expose template

3. Add radioactive primer

4. Add 90% dNTPs

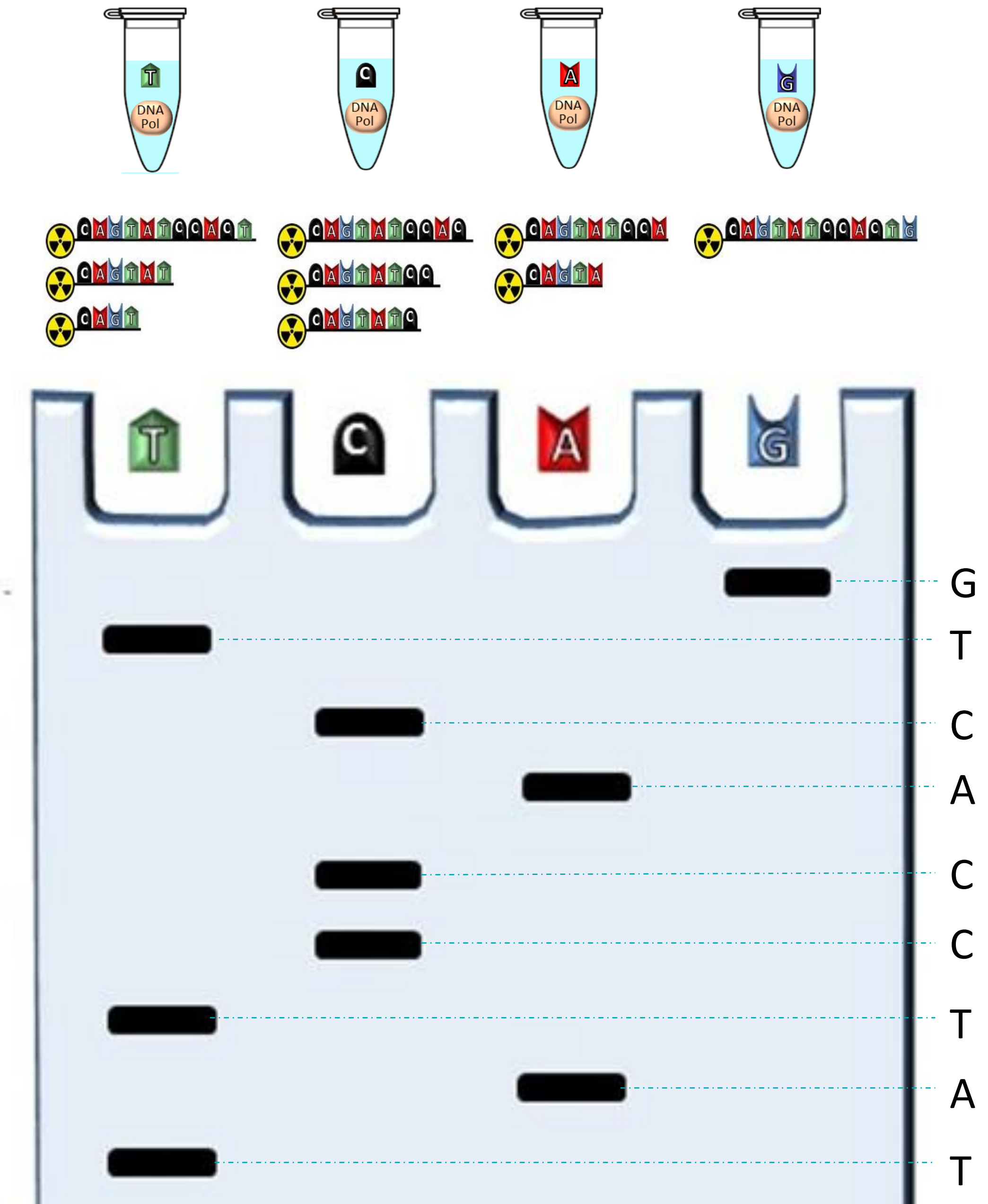
5. Make 4 aliquots



6. Add 10% dideoxy NTPs

7. Add Enzyme

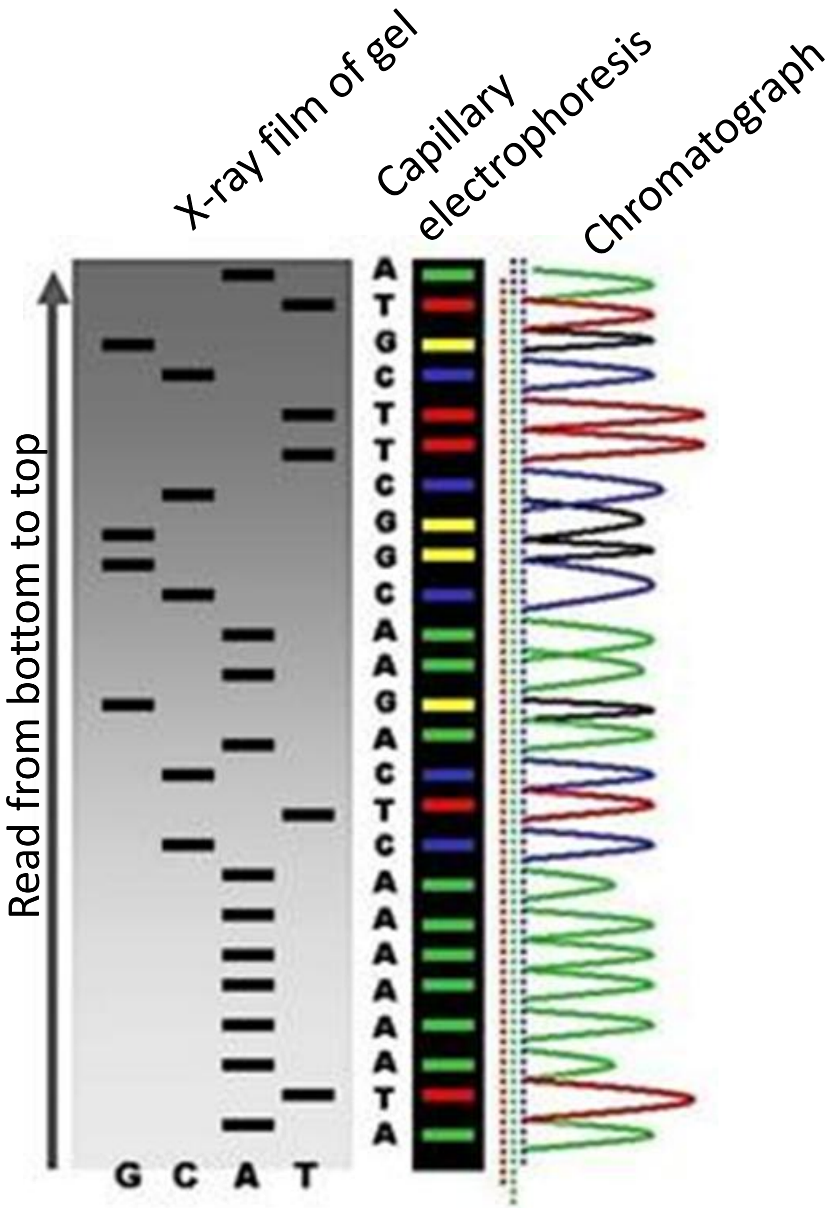
Sanger Sequencing: Sequencing Reaction Run on Gel



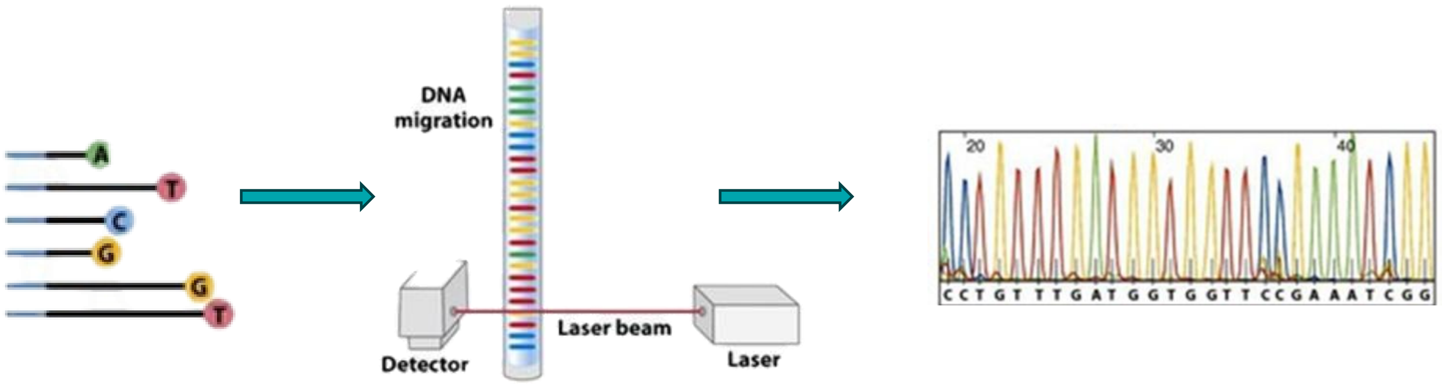
8. Synthesis of complimentary strand occurs in each tube.
9. Random incorporation of ddNTP and synthesis is then terminated
10. Conditions set with substrates in excess
11. Thousands of fragments made, each terminated at a different base pair
12. Reaction is loaded onto a gel
13. Fragments are separated according to size
14. Gel is exposed to X-ray film
15. Weight of fragments infers sequence
16. Sequence read from bottom to top

Original DNA

Sanger Sequencing – Automated with fluorescence



- Same set up as the original, with exceptions:
  - ddNTPs are labeled uniquely with fluorescent tag
  - All in one reaction tube
  - Whole reaction applied to capillary electrophoresis
  - Sequence read by computer to generate chromatogram



Expression of a Gene

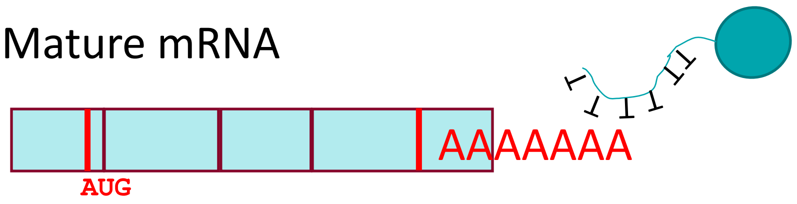
RNA- Northern blot

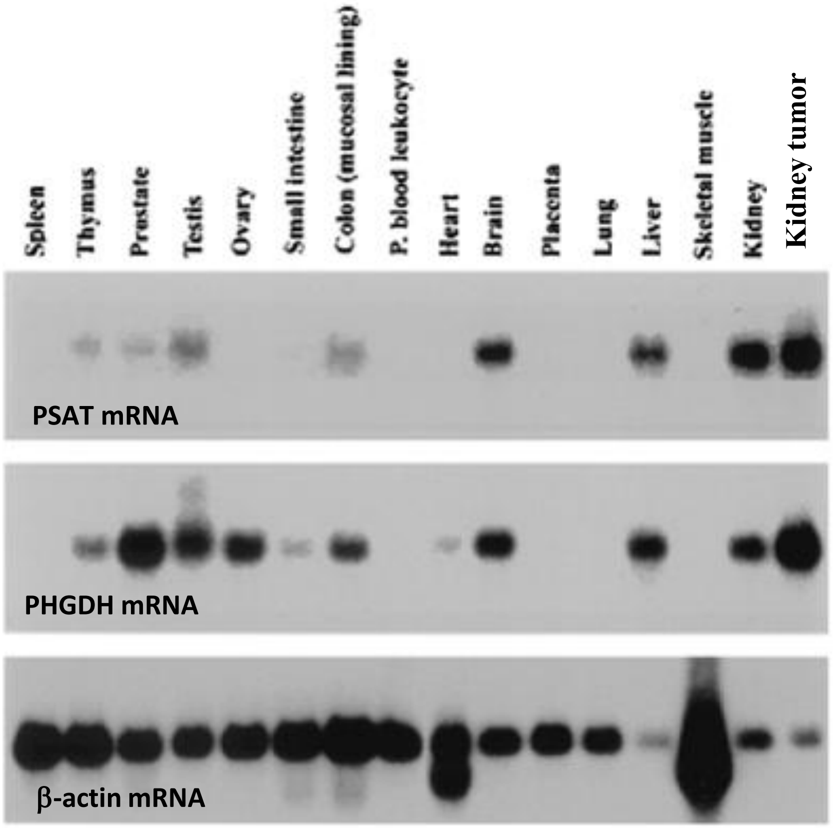
Protein-Western Blot

RNA – Northern Blot

- Northern blot: method similar to Southern blot
- **mRNA** is isolated from cells & separated by gel electrophoresis (Does RNA have a charge?)
- Transferred to a membrane
- Hybridized with specific ssDNA probe, radiolabeled
- Excess probe washed off; membrane dried & exposed to x-ray film
- Detects amount and size of the target mRNA
- Excellent for detecting the expression of a gene in different tissues

Mature mRNA

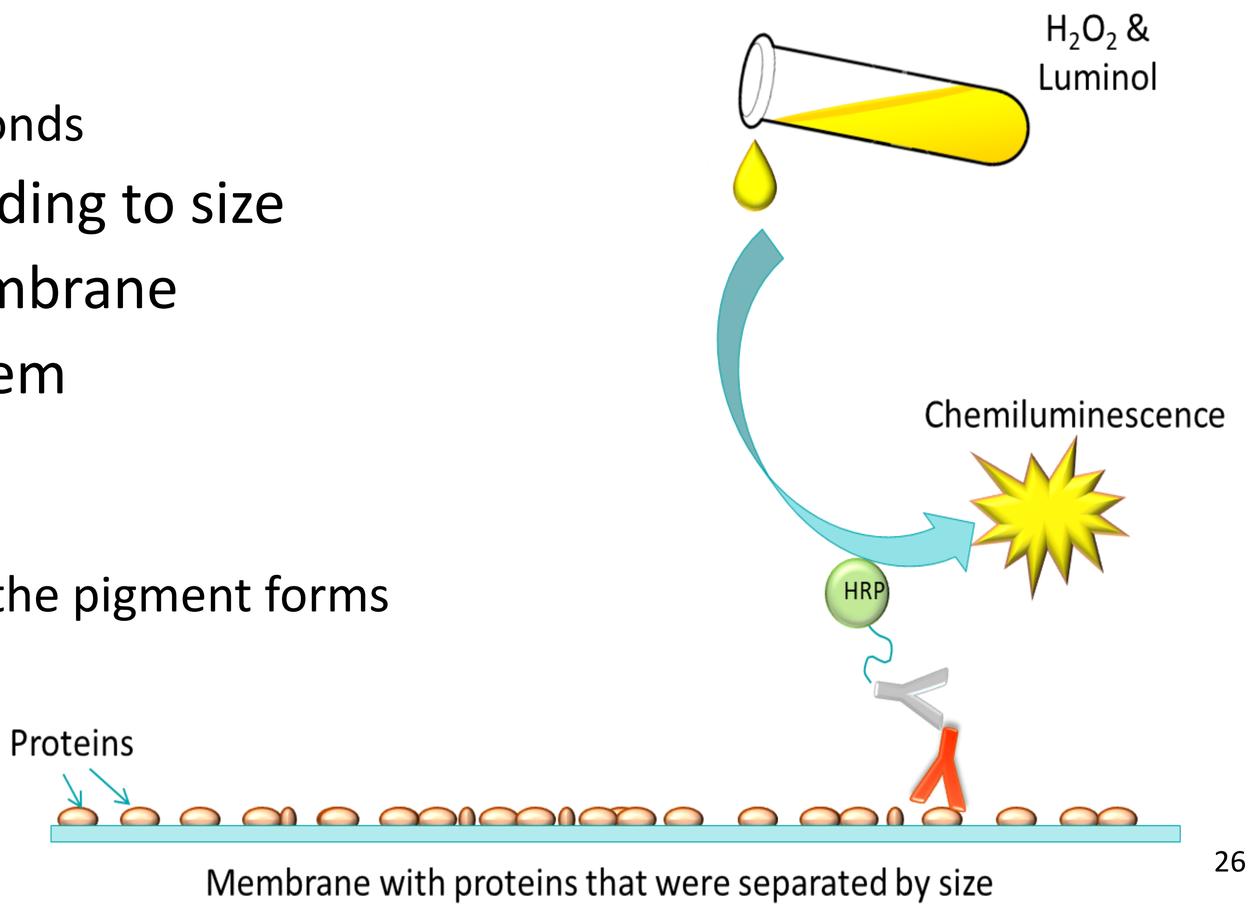




Slightly different sizes indicate splice variants

Protein: Western blots

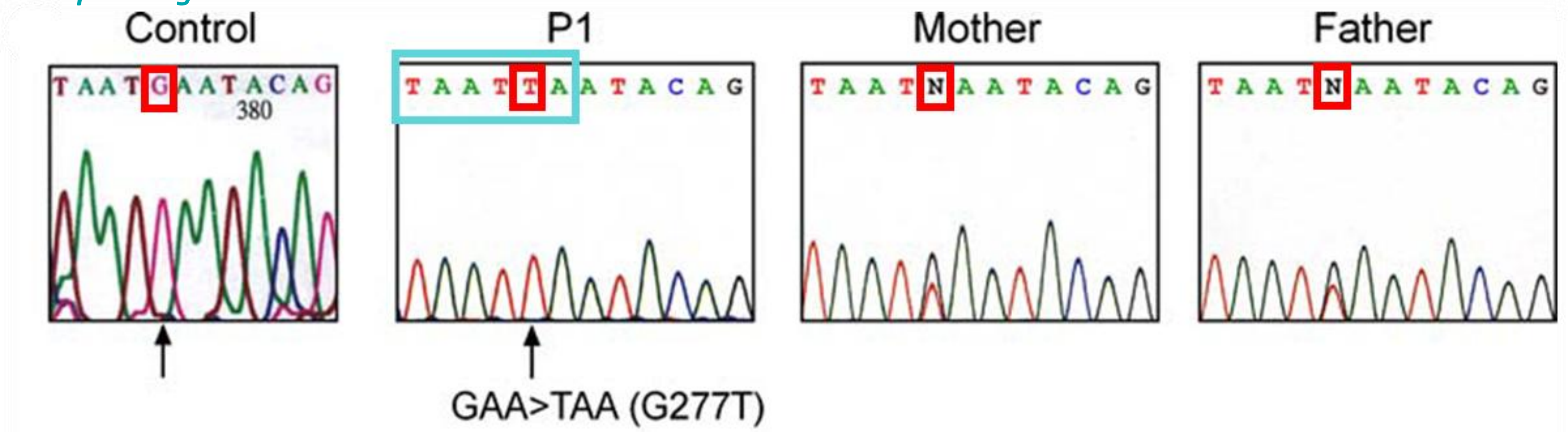
- Proteins are isolated from cells
- Purified proteins denatured with heat and detergent
  - Peptide is coated with negatively charged SDS detergent
- Disulfide bonds broken
  - A reducing agent used to break disulfide bonds
- Electrophoresis separate proteins according to size
- Transfer peptides from gel to nylon membrane
- Detection using a labeled antibody system
  - Visualization using chemiluminescence,
    - Treated membrane is exposed to x-ray film
  - Visualization using chromogenic substrate the pigment forms directly on the membrane



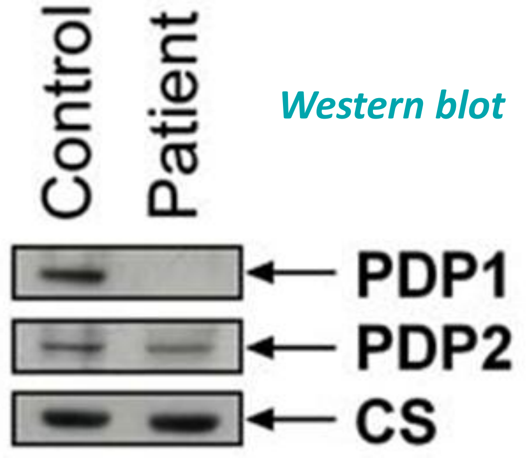
Does protein have a charge??

Compare data for PDP1 mutation: Sequencing, RFLP, & Western

Sequencing



Western blot



- Infantile lactic acidemia, low pyruvate dehydrogenase complex (PDH) activity, no mutations found in any complex subunits
- Sequencing showed a mutation in an enzyme that activates PDH complex → Pyruvate dehydrogenase phosphatase (PDP1)
- Mutation creates a restriction site and premature stop codon
- Antibodies made against recombinant PDP1 and PDP2 proteins
- "Private" mutation, specific to this family and not common

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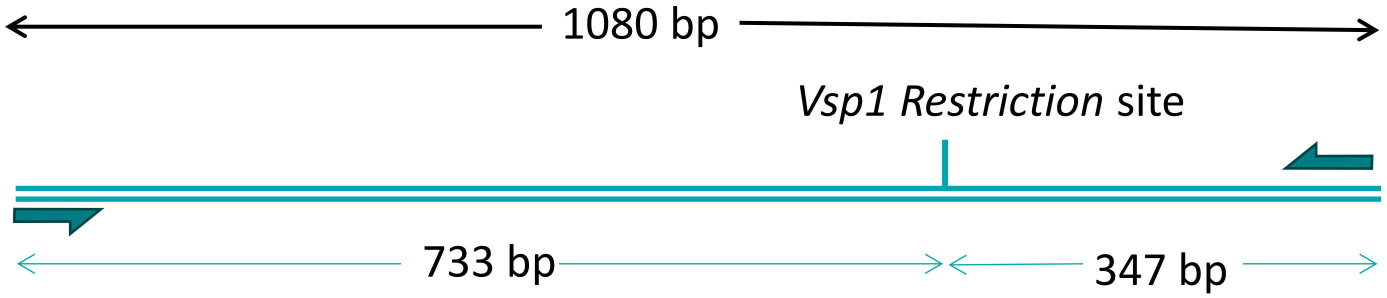
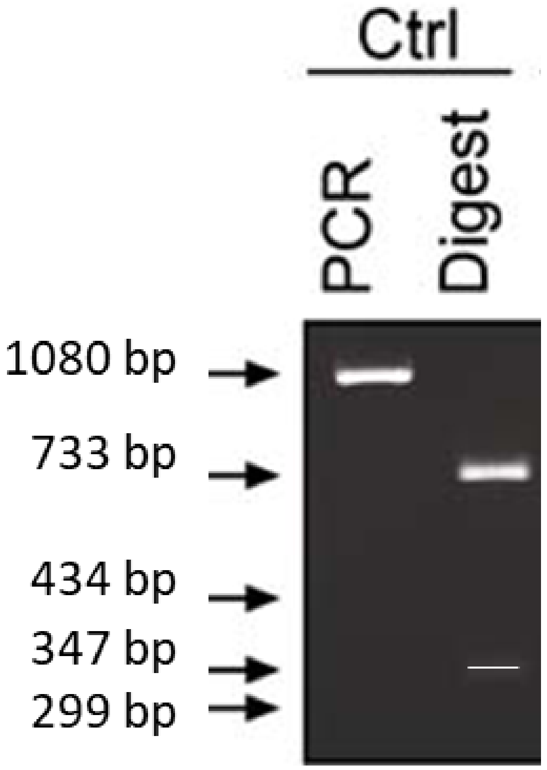
Compare data for PDP1 Mutation: Here is the test: Control

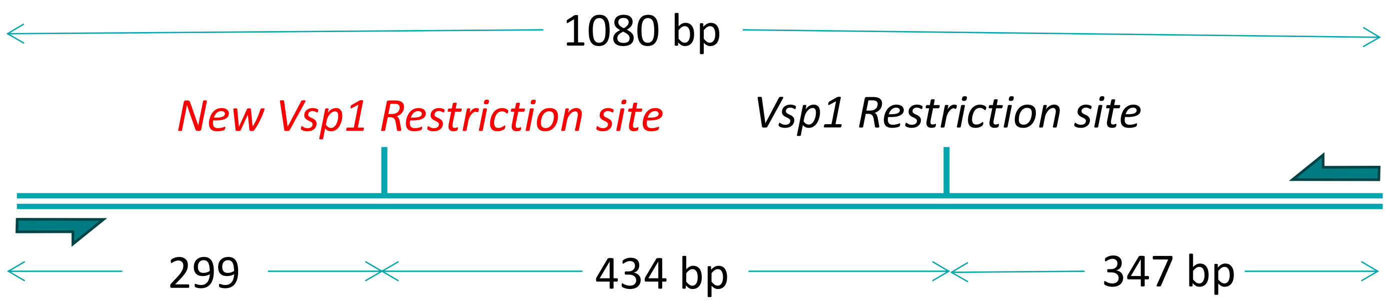
- PCR 1084 bp fragment
- **Normal Allele**, the 1084 bp fragment should only have one *Vsp*I site (the restriction site)
- After digestion, two fragments 733 & 347 bp

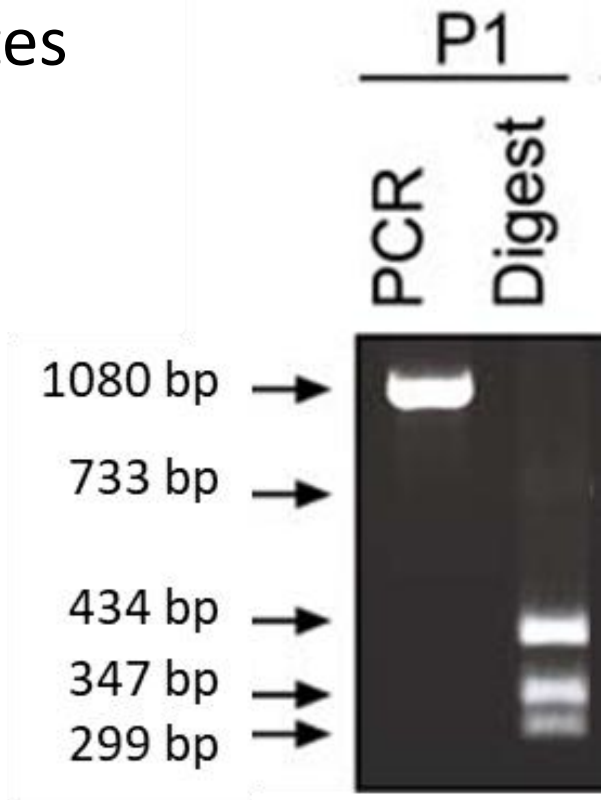




Compare data for PDP1 Mutation: Here is the patient

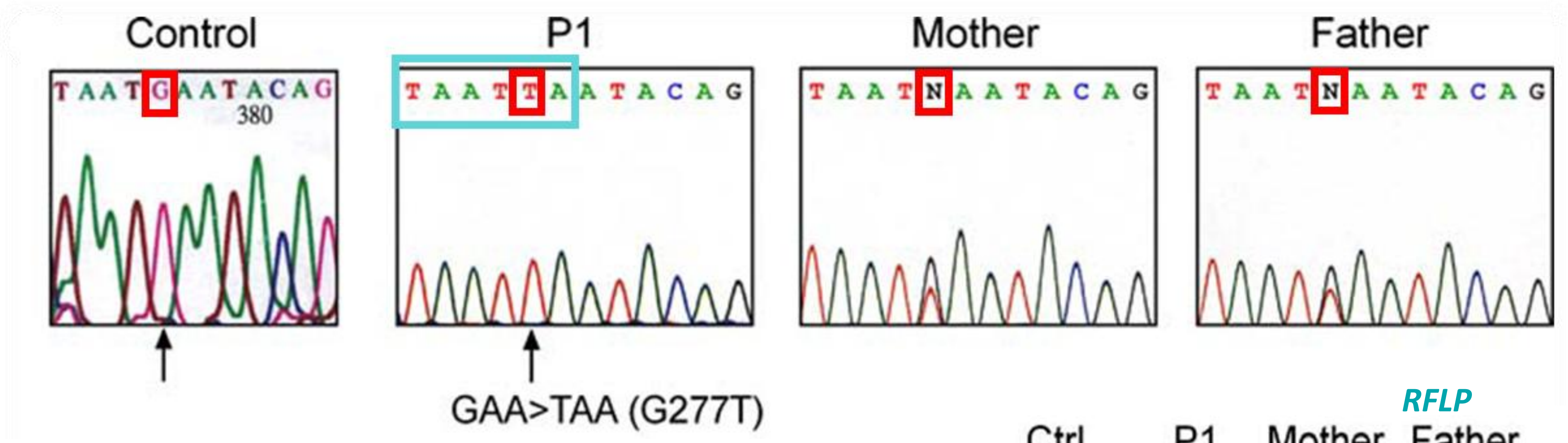
- PCR 1084 bp fragment
- **Mutant Allele**, the 1084 bp fragment has two VspI sites
- After digestion, 3 fragments 299, 434 & 347 bp



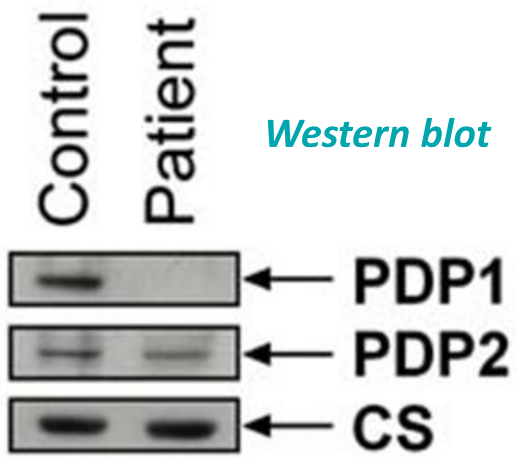


Compare data for PDP1 mutation: Sequencing, RFLP, & Western

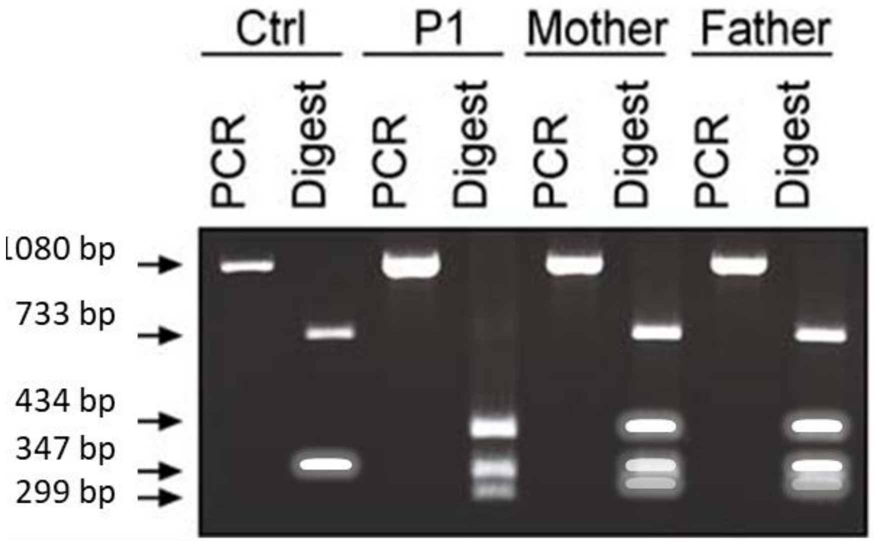
Sequencing



Western blot



RFLP



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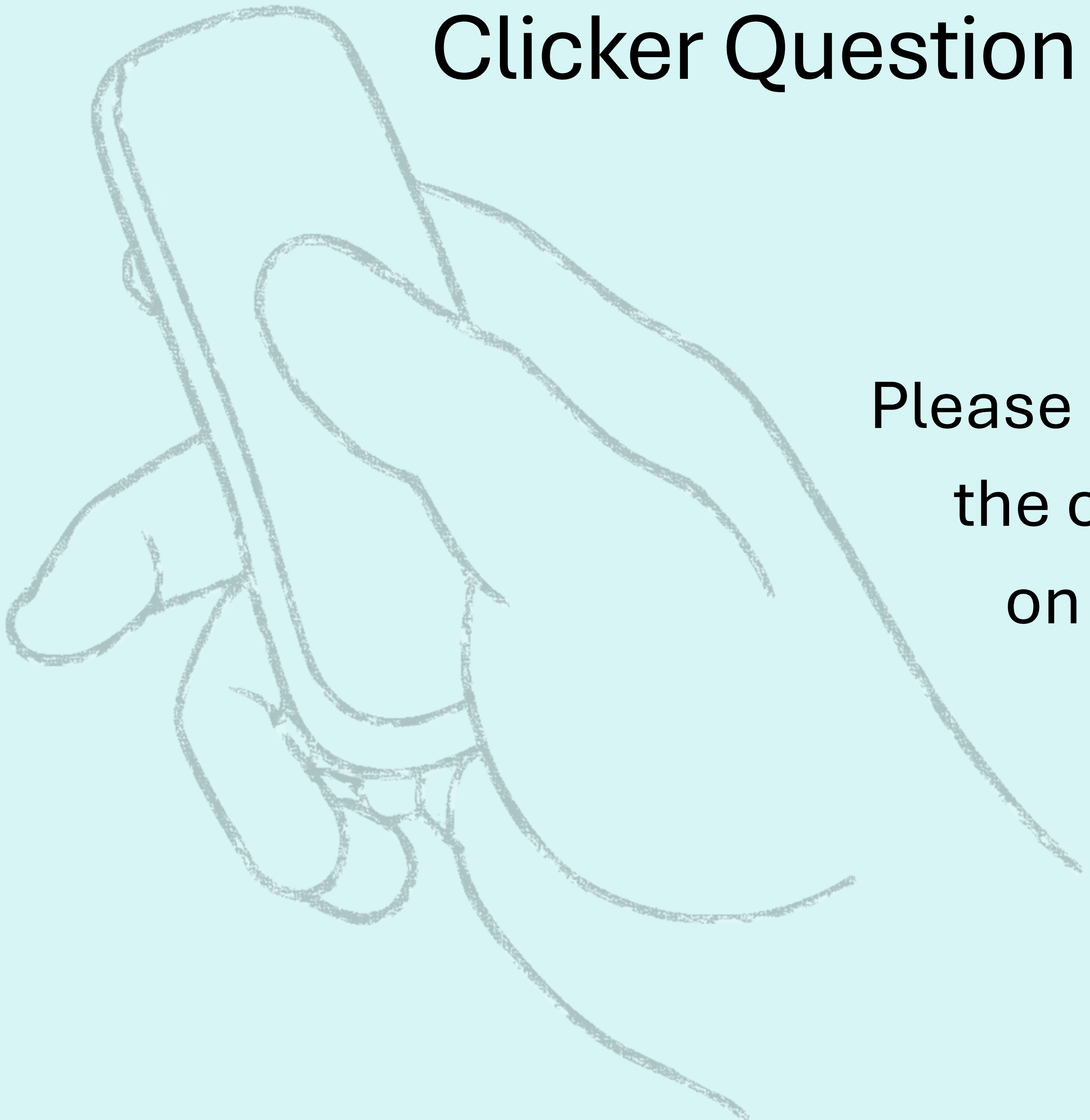
Clicker Question Next!



Please prepare to answer
the clicker question
on the next slide

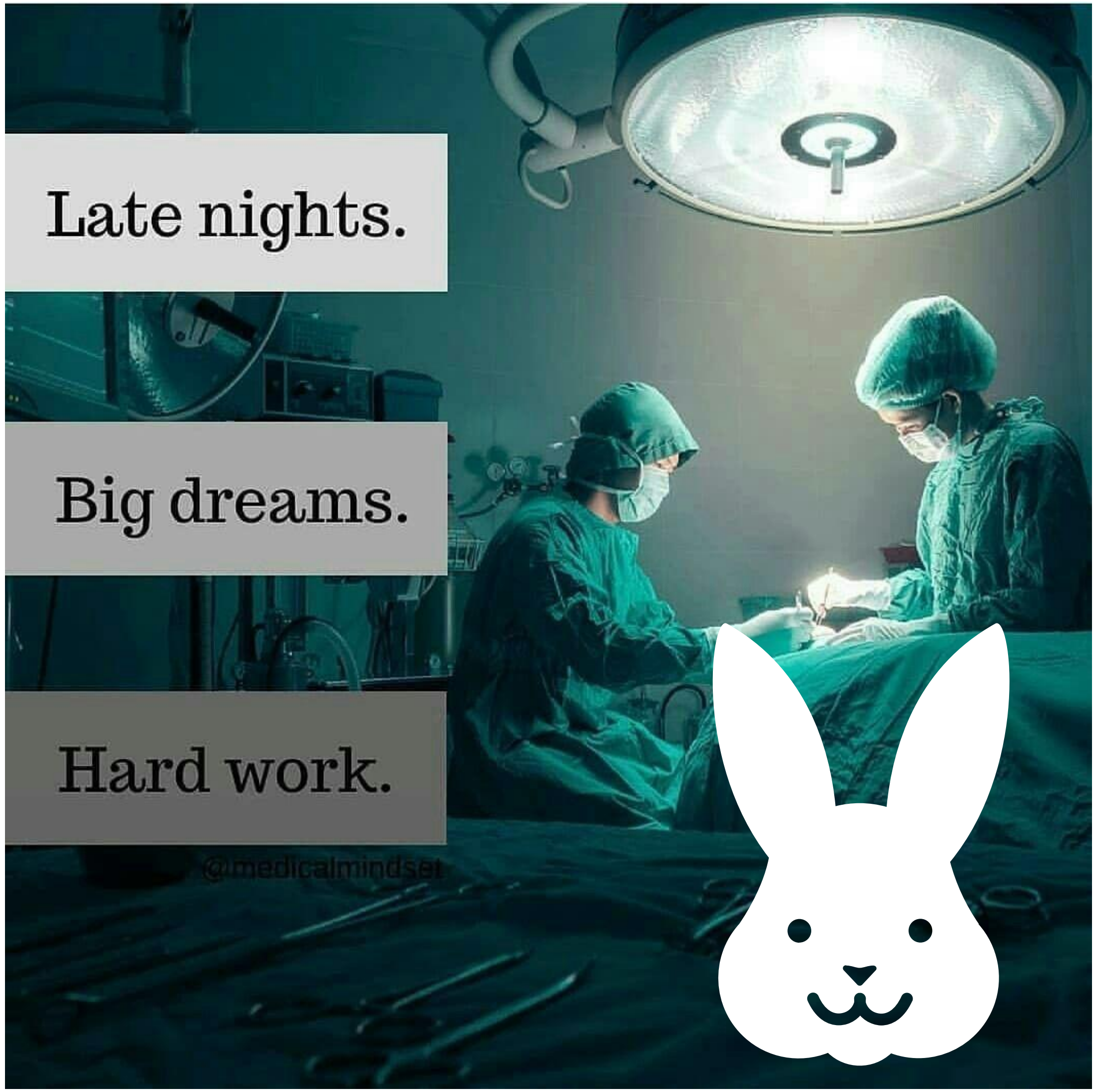
Summary for self study

Test	Method	Application
Restriction enzyme digest Sometimes referred to as restriction endonuclease	Genomic DNA is isolated from cells and digested with restriction enzymes which cuts the DNA in a reproducible manner, easier to work with the fragments	Cut DNA and separate fragments by gel electrophoresis. Fragment of interest that changes size could be due to insertion, deletion, or may be completely missing. Southern blotting, RFLP analysis, cloning
DNA cloning – unknown DNA	Target genomic DNA is cut by restriction endonuclease (RE), fragments are ligated into a DNA vector that has been cut with the same RE. When the DNA fragment is ligated into a DNA vector, it is called recombinant. Then, recombinant vector is inserted into a cell, the vector (sometimes called plasmid) will multiply many times in cell to amplify the DNA	The recombinant vector can be specialized to: A. Divide many times in the cell to make lots of DNA of interest B. Can have special region before the inserted DNA where a universal sequencing primer will attach for Sanger sequencing
DNA cloning – known DNA	Target DNA is usually thought to contain a mutation in the gene. A. An appropriate RE is used to cut the gene from genomic DNA to amplify the amount of DNA B. mRNA of interest is converted to cDNA which is ligated into an expression vector	A. DNA of a gene is amplified to generate DNA for Sanger sequencing to look for mutation, or other investigations where you need lots of specific DNA B. cDNA is ligated into an expression vector to make a recombinant protein (antibody generation & biochemical assays)
cDNA generation - RT PCR (reverse transcriptase PCR where both steps occur in the PDR machine)	mRNA is isolated from a cell and **r**everse **t**ranscriptase is used to generate complimentary copy of the mRNA. DNA polymerase is used as the second step to complete the generation of double stranded DNA from the mRNA. Whole region is the open reading frame of a gene and PCR makes millions of copies	A. Expression analysis of all the mRNA expressed in a cell B. Take the cDNA which codes for a protein of interest is put into an expression vector to make recombinant protein
Probe generation	-ssDNA probe is designed to hybridize DNA or RNA of interest, but must have some type of tag to identify where it has hybridized, either radioactive or fluorescent -Antibody made against protein	-ssDNA-Southern blot (all 3 types), Northern blot, ASO, PCR primer, sequencing primer -Antibody-Western blot

Test	Method	Application
Southern blotting-insertion or deletion of DNA	Genomic DNA is fragmented with RE, fragments are separated by size by gel electrophoresis, fragments transferred to membrane, probe designed to hybridize to fragment of interest, after hybridization-washing-drying membrane, expose to X-ray film to interpret results	Looking for insertions or deletions in pre-determined regions of interest Still used if the region of DNA is difficult to amplify by PCR (GC rich regions, highly repeated regions, inverted repeat regions)
Southern blotting-RFLP	Regular Southern blot protocol but the RE used can identify mutations that will make or break a restriction site within the target DNA	Quickly identifying a mutation that makes or breaks a restriction site Still used if the region of DNA is difficult to amplify by PCR (GC rich regions, highly repeated regions, inverted repeat regions)
Southern blotting-Methylation	Regular Southern blot protocol but using a RE that targets in a region that may be methylated Done twice, before and after DNA is treated with an enzyme that removes DNA methylation	Fragmentation pattern is analyzed before and after DNA has been treated with methylase Still used as methylation pattern is only retained in-vivo, many cells must be used to get sufficient genomic DNA for testing
PCR	DNA is isolated from cell, primers made to amplify DNA of interest (primer pair, one that binds to the top strand and one that binds to the bottom strand). About 30 cycles are used to amplify DNA of interest. Cycle: 1. denature genomic DNA with heat 2. reduce the temperature to allow primers to bind 3. reduce the temperature again to allow thermostable DNA polymerase to synthesize 2 daughter strands of DNA (one from top strand and the other from bottom strand)	Used to amplify DNA for further testing if a region of DNA must be enriched. Examples: forensic analysis of a small biological sample, RFLP without using Southern blot
PCR-RFLP	Amplification of genomic DNA of interest that may have a mutation that makes or breaks a restriction site, then take the sample and add the restriction endonuclease that will digest the restriction site of interest	Amplification of genomic DNA of interest that may have a mutation that makes or breaks a restriction site. About 30% of pathogenic mutations in genes will make or break a restriction site, and this is an easy test to develop for **familial mutations**, not common in the general population
ASO	Genomic DNA is isolated and fixed on a membrane at two spots. Two probes are designed, one designed to specifically hybridize to the normal allele and the other to specifically hybridize to the mutant allele. Probes are tagged with radioactivity or fluorescence. Membrane is exposed to x-ray film to determine if the probe hybridized to an allele. If it binds, then the patient has that allele. If no hybridization occurs, the patient does not have that allele	Designed for **common** mutations in the general population. It is a very stringent test, as probes are 15-30 bp long and will not hybridize if even only one base pair is different. Takes a lot of work up to make it a diagnostically relevant test. Panels of ASO tests are now offered, testing the most common mutations found in the general population

Test	Method	Application
Gene expression – North blot	mRNA is isolated from cells or tissues of interest, and separated by size by gel electrophoresis, mRNA in gel is transferred to a membrane, a specific probe is applied to hybridize to an mRNA of interest, probe is tagged with radioactivity or fluorescence, membrane is washed and dried, and exposed to x-ray film to detect the presence of mRNA of interest	A. mRNA levels in different tissues can be analyzed B. mRNA levels can be compared between healthy & diseased tissues to determine if disease alters the expression of a gene C. mRNA size may be changed of mutation affected splice sites D. mRNA may be absent if mutation occurred in the promoter and RNA polymerase is unable to bind template
Gene expression – Western blot	Proteins are isolated from cytosol, denatured, coated with detergent to mask charges of amino acid side groups, di-sulfur bonds broken with a reducing agent, and proteins are separated by size by gel electrophoresis. Proteins are transferred from gel to a membrane, and an antibody targeting protein of interest is applied to the membrane. A second antibody, designed to bind to the first antibody, has an enzyme attached to it which will activate chemiluminescence if the protein of interest was in the sample. Usually, the first antibody made against protein of interest in rabbit, second antibody is made against rabbit antibodies in a goat	Determine protein levels in the cell. Some mutations do not change the size of DNA or mRNA but may affect the protein. Examples: A. Premature stop codon mutation in mRNA that is recognized by ribosome B. Base pair deletion or insertion which changes the reading frame of a gene so that the amino acids are all different and no longer recognized by the antibody
Sanger sequencing – the original with radioactive primer	Genomic DNA is isolated & denatured to allow a primer to bind (with radioactive tag). Substrate of all deoxynucleotides are added (all 4 dNTPs). The solution is divided into 4 tubes. Each tube has a 10% amount of a unique di-dioxy nucleotide (missing the all important 3'OH required for elongation of the growing complimentary strand). DNA polymerase is added, and DNA synthesis will occur. The dideoxy nucleotide will randomly be inserted into the growing chain by DNA polymerase, but once it is added, synthesis is terminated. The reaction tube has millions of fragments of synthesized DNA, and the protocol is optimized to assure that there are enough fragments generated of different sizes to represent the sequence of the template. The reactions are applied to a vertical gel for electrophoresis, fragments separated by size. Gel is fixed to a membrane, dried and exposed to x-ray film. The film will show the many fragments, which should reveal the sequence when read from bottom to top.	This is still used if sequencing genomic DNA which does not have a reference genome. Sequencing here is done in a step-wise fashion where the information from sequencing provides the next primer to be used to give further information of the genome
Sanger sequencing- with dideoxy nucleotides, tagged with unique fluorescent color	DNA of interest is typically ligated into a sequencing vector or artificial chromosome which has a universal sequencing primer binding site. One reaction tube is made, containing all ddNTs with unique fluorescent tag, and separated by size with capillary electrophoresis. Laser, detector and computer will give a printout of results called chromatograph	This is most popular in research labs to identify novel mutations in a target gene, can also confirm cDNA sequence that was ligated in a cloning or expression vector. Now, a mutation is often verified by next generation sequencing with in-depth read.


IF YOU WANT TO BE SUCCESSFUL,
YOU HAVE TO BE WILLING TO
DISAPPEAR FOR A WHILE.


Late nights.
Big dreams.
Hard work.
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The End